CCR-k Arithmetic Refinement:
p-Adic Resonance, Ramification, and Arithmetic Density Laws

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Abstract

We study the p-adic arithmetic of stationary normalized columns in the previously established higher-order chronological carry refinement (CCR-k) hierarchy. The organizing invariant is an exact first-order Stirling source obstruction. At every odd order $r \geq 5$, its integer-source image is the explicit lattice $1 + d_r\mathbb{Z}$, where d_r has a closed prime-factor formula and $12 \mid d_r$. This rules out zero-source resonance at all but finitely many primes for each fixed integral source vector, whereas Haar-varying p-adic source coordinates obey independent quadratic-fold laws. We establish finite-precision congruence moments, a multicolumn product law, and an adelic prime/height dichotomy. The arithmetic phenomena are made explicit in a complete fourth-column local factorization, including a terminating 419-adic integer discriminant certificate, and in the exceptional fifth column at $p = 73$, where two critical folds admit exact covariance and joint-density formulas (including five-root density $1/21904$). Further specializations give seventh- and ninth-column prime-axis Galois laws. We also distinguish stationary source cancellation from time-dependent integer-source observations: exact multitime congruence and nonnegative semigroup criteria, with a sharp signed Hessenberg-band threshold, determine which terminal scalar shadows are attainable. All claims concern the explicitly typed CCR polynomial and raw-source models, not arbitrary physically realizable pushdown dynamics. The Nörlund, local-field, Galois, numerical-semigroup and probabilistic methods used are classical; the proposed contribution is their rigorously delimited synthesis driven by the CCR source deformation.

Keywords: chronological carry refinement; p-adic polynomial roots; Newton polygons; exceptional primes; discriminants; Chebotarev density; Hensel lifting.
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1 Introduction and provenance

Chronological carry refinement (CCR-k) admits a typed higher-order transport and universal normalized columns; these originated in the preceding CCR-3, CCR-4, and CCR-k developments [1, 2, 3]. The point of the present work is not to reprove that transport or to introduce a fourth-order CCR-4 model. Throughout, *fourth column* means the index $r = 4$ within the full CCR-k theory, valid for any $k \geq 4$. We treat the associated polynomial as a formal arithmetic object, without asserting that every algebraic parameter has a physically realized chronological DPDA source.

Main contributions and scope. Beyond the previously established typed transport, the results proved here form four connected layers: (i) explicit arithmetic classification of the

normalized fourth- and fifth-column polynomials, with exceptional-prime certificates at 419 and 73; (ii) the all-odd-order Stirling source obstruction, its exact integral lattice and its finite support in the prime axis; (iii) multicolumn and adelic Haar laws, carefully separated from fixed-integer source quantifiers; and (iv) a precise boundary between stationary source invariants and time-dependent terminal raw-coefficient observations. The novelty being proposed is *the combination and exact specialization to CCR source transport*, not a claim to have invented Hensel lifting, Nörlund polynomials, the Maxwell/caustic factorization, generic two-coefficient Galois theory, or numerical-semigroup methods. For the latter antecedents see, respectively, [6, 7, 4, 16, 17, 18, 19, 21].

The starting quartic belongs to the preceding CCR development; thus *the polynomial formula is prior work*. The material developed here is its all-prime local root-and-ramification stratification, a discriminant-level finite certificate, and the two complementary prime/parameter counting laws, including a precision-level moment theory. In the spectral background, the original Hessenberg principal-determinant polynomials Q_n must not be confused with the fixed normalized column polynomial $\Pi_{p,4}^{(\lambda)}$.

1.1 Stationary transport conventions and arithmetic scope

For clarity we record the stationary normalized system used throughout. Write $\ell_m = m + p(m-1)$, set $\lambda_3 = 1$, and take the order-a stationary source to be $a_{a,m} = (-1)^a p\lambda_a$ $(a \geq 3)$, independent of m. In the integral-source model, $\lambda_a \in \mathbb{Z}$; where explicitly stated, a local Haar model instead varies the relevant coordinates in $\mathbb{Z}_p$. With $\Pi_{p,0} = 1$ and with terms of negative column index omitted, the established normalized fixed-column transport is

$$\begin{aligned} T_j\Pi_{p,j}(m) &= \ell_m \Pi_{p,j-1}(m) + p\ell_{m-1}\Pi_{p,j-2}(m-1) \\ &\quad + p\sum_{a=3}^{j}(-1)^a\lambda_a\,\Pi_{p,j-a}(m-a+1), \qquad (1) \\ (T_j f)(m) &= mf(m+1) - (m-j)f(m). \end{aligned}$$

For actual normalized rows this transport is imposed for $m \geq j+1$, with the first admissible value fixed by the separate boundary relation at $m = j$; its unique polynomial continuation is the object used in formal specialization at $p = 0$. All these conventions and polynomiality follow from the preceding CCR-k theory [3]; they are not new existence claims. In residue calculations we either use an integral numerator or require the relevant denominators to be p-adic units. The special case $p = 5$ of the quartic below uses its integral numerator.

1.2 Notation and the established quartic

For a prime $p \geq 5$ and a stationary integer parameter λ, put

$$\begin{aligned} G_{p,\lambda}(m) &= 15(p+1)^4m^4 + 30(p+1)^3(3p-1)m^3 \\ &\quad - 5(p+1)^2(287p^2+158p-1)m^2 \\ &\quad + 2(p+1)^2(1645p^2-50p+1)m \\ &\quad + 240p(6-5p^3-5p^2+11p+5) + 1440p(\lambda-1). \qquad (2) \end{aligned}$$

The normalized column is $\Pi_{p,4}^{(\lambda)} = G_{p,\lambda}/5760$; its roots in $\mathbb{Q}_p$ are those of $G_{p,\lambda}$. Crucially, at $p = 5$ one works with the *integral numerator* G, not a coefficientwise reduction of Π. Write

$$H(m) = 15m^3 - 30m^2 + 5m + 2, \qquad G_{p,\lambda}(m) \equiv mH(m) \pmod{p}. \qquad (3)$$

The parameter λ disappears in this reduction. Direct computation gives

$$\operatorname{disc}(H) = 125700 = 10^2 \cdot 3 \cdot 419, \qquad \operatorname{disc}(mH) = 502800 = 2^4 \cdot 3 \cdot 5^2 \cdot 419. \qquad (4)$$

2 The complete local factorization classification

Theorem 2.1 (All-prime root and factor stratification)**.** *For each integer λ and prime $p \geq 5$:*

1. *If $p \geq 7$ and $p \neq 419$, the roots of $G_{p,\lambda}$ in $\mathbb{Q}_p$ are precisely the unique lifts of the roots of $mH(m)$ in $\mathbb{F}_p$. Their number is 1, 2, or 4, independent of λ, and all residual irreducible factors lift unramified.*

2. *If $p = 5$, there is exactly one root in $\mathbb{Q}_5$. For $\lambda \neq 115$ its valuation is $2 + v_5(\lambda - 115)$, and at $\lambda = 115$ the root is 0. The remaining degree-three factor is irreducible and totally ramified over $\mathbb{Q}_5$.*

3. *If $p = 419$, the simple residues $m \equiv 0, 20 \pmod{419}$ always lift to two distinct roots. A further quadratic pair in $m \equiv -9 \pmod{419}$ is controlled by a unique 419-adic critical parameter Λ. Its exact splitting and ramification rule is stated in Theorem 4.1 below for all integer λ.*

Proof. For $p \geq 7$, $p \neq 419$, the coefficient of m^4 in G is a unit; the reduction mH is squarefree by (4) (and $H(0) = 2$). Ordinary Hensel lifting gives a bijection of linear residue factors with $\mathbb{Q}_p$-roots. Hensel factorization also lifts each separable irreducible factor of the reduction, yielding unramified extensions.

At $p = 5$, direct specialization is

$$G_{5,\lambda}(m) = 19440m^4 + 90720m^3 - 1433520m^2 + 2943072m + 7200(\lambda - 115).$$

The coefficients of m, m^2, m^3, m^4 have valuations $0, 1, 1, 1$, respectively. For $\lambda \neq 115$ the constant term has valuation $2 + d$, where $d = v_5(\lambda - 115)$. Its Newton polygon has segments $(0, 2+d) \to (1, 0) \to (4, 1)$: the first slope gives one integral root of valuation $2+d$; the second gives three conjugate roots of valuation $-1/3$. A degree-three factor whose root valuation has denominator 3 is totally ramified and irreducible. When $\lambda = 115$, factor out m and apply the same final segment to the cubic.

At $p = 419$, one checks the residue factorization

$$G_{419,\lambda}(m) \equiv 15m(m - 20)(m + 9)^2 \pmod{419}.$$

The two simple residues lift. The double residue is analyzed below with the stationary derivative and a p-adic quadratic normal form. □

3 The 419-adic critical coordinate

Let $f_\lambda(m) = G_{419,\lambda}(m)$. Its derivative is independent of λ and satisfies

$$f'_\lambda(-9) \equiv 0, \qquad \frac{1}{2} f''_\lambda(-9) \equiv 144 \pmod{419}.$$

Hensel's lemma yields a unique $c \in \mathbb{Z}_{419}$ such that $c \equiv -9 \pmod{419}$ and $f'_\lambda(c) = 0$. Since the parameter occurs as $1440 \cdot 419\lambda$, there is a unique $\Lambda \in \mathbb{Z}_{419}$ satisfying $f_\Lambda(c) = 0$.

Lemma 3.1 (Exact quadratic normal form)**.** *There is an analytic invertible change of variable near c, defined over $\mathbb{Z}_{419}$, under which the quadratic branch of $f_\lambda(m) = 0$ becomes*

$$z^2 = -10 \cdot 419\,(\lambda - \Lambda). \tag{5}$$

For $\lambda \neq \Lambda$, put $h = v_{419}(\lambda - \Lambda)$. The quadratic is totally ramified if h is even; if h is odd, it splits over $\mathbb{Q}_{419}$ exactly when the leading unit of $-10(\lambda - \Lambda)/419^h$ is a square modulo 419, and otherwise it is an unramified irreducible quadratic.

Proof. For $m = c + y$, stationary Taylor expansion yields

$$f_\lambda(c+y) = 1440 \cdot 419(\lambda - \Lambda) + y^2 U(y), \quad U(0) \equiv 144 \pmod{419}.$$

For $y \in 419\mathbb{Z}_{419}$, the unit $U(y)/144$ is congruent to 1 modulo 419. Its unique square root congruent to 1 defines the analytic coordinate $z = y\sqrt{U(y)/144}$; moreover it is an isometry of the entire residue disk, not merely a local map with unit derivative. Indeed, for $s(y) = \sqrt{U(y)/144}$, the integral convergent power series on $419\mathbb{Z}_{419}$ satisfies $s(y) \equiv 1 \pmod{419}$ and $s(y) - s(y') \in (y - y')\mathbb{Z}_{419}$. Thus

$$z(y) - z(y') = (y - y') \left(s(y) + y' \frac{s(y) - s(y')}{y - y'} \right),$$

and the factor in parentheses is congruent to 1 modulo 419. Hensel lifting from $y = z = 0 \pmod{419}$ gives surjectivity and a unique inverse modulo every 419^n, hence a bi-analytic disk isometry. It gives (5) exactly on that disk, and therefore all integral-root and finite-congruence assertions. In the nonresonant case $v_{419}(\lambda - \Lambda) = 0$, however, the two algebraic zeros have $v_{419}(y) = 1/2$ and *lie outside* the integral residue disk: their irreducibility and total ramification follow separately from the translated polynomial's Newton segment joining $(0, 1)$ to $(2, 0)$, using $f'(c) = 0$ and $U(0)$ a unit. The same Newton-polygon argument handles all odd critical-value valuations; even critical-value valuations give the split/unramified cases by the local square-unit criterion. This separation of the integral disk from extension-valued roots is essential, and is consistent with the established p-adic Morse/inverse-function framework [12]. □

4 An exact integer discriminant certificate

The preceding description refers to the non-rational critical coordinate Λ. Our next result eliminates it from every integer-parameter decision.

$$\boxed{\begin{aligned} F(T) = {} & 127983969T^3 - 773766474332943627T^2 \\ & + 19464840642527447214601687T \\ & + 25531064848926331469188878661677 79. \end{aligned}} \tag{6}$$

Theorem 4.1 (Integer discriminant certificate and effective resonance bound)**.** *The polynomial $F(T) \in \mathbb{Z}[T]$ is irreducible over $\mathbb{Q}$, and its unique zero in $\mathbb{Z}_{419}$ is Λ. Consequently, $\Lambda \notin \mathbb{Q}$; in particular, $\lambda \neq \Lambda$ for every integer λ.*

For any $\lambda \in \mathbb{Z}$, calculate the nonzero integer *$F(\lambda)$ and set*

$$k = v_{419}(F(\lambda)), \qquad u = 419^{-k} F(\lambda) \pmod{419} \in \mathbb{F}_{419}^\times.$$

The full local factorization of $G_{419,\lambda}$ is given by the following finite certificate:

Criterion	*Additional quadratic factor*	$\mathbb{Q}_{419}$*-roots*
k even	*irreducible, totally ramified*	2
k odd, $\left(\frac{374u}{419}\right) = +1$	*splits into linear factors*	4
k odd, $\left(\frac{374u}{419}\right) = -1$	*irreducible, unramified*	2

In particular $G_{419,\lambda}$ is separable for every integer λ and the exceptional repeated-root branch of Lemma 3.1 is not attained by integer sources. *Moreover*

$$k \leq \log_{419} |F(\lambda)| \leq \log_{419} C + 3 \log_{419}(1 + |\lambda|), \quad C = \sum_{j=0}^{3} |[T^j]F|. \tag{7}$$

This is an explicit input-height bound on resonance depth and a terminating integral decision algorithm.

Proof. Independently calculate the discriminant of the quartic (2) with $p = 419$ as a polynomial in the parameter:

$$\operatorname{disc}_m G_{419,T} = C_0 F(T), \quad C_0 = 44676505705166237495036215296000000000000000000 \neq 0. \tag{8}$$

Modulo 23, we have

$$F(T) \equiv 9(T^3 + 5T^2 + 2T - 6) \pmod{23}.$$

The displayed cubic has no root in $\mathbb{F}_{23}$; as its degree is three, it is irreducible. Thus F is irreducible in $\mathbb{Q}[T]$. Modulo 419,

$$F(T) \equiv 233(T - 132) \pmod{419},$$

and 233 is a unit. Hensel gives a unique zero $\Lambda_0 \in \mathbb{Z}_{419}$. But $G_{419,\Lambda}$ has a multiple root c by construction, so (8) implies $F(\Lambda) = 0$. Therefore $\Lambda_0 = \Lambda$.

The exact difference identity

$$F(T) - F(\Lambda) = (T - \Lambda)W(T, \Lambda), \qquad W(T, \Lambda) \equiv 233 \pmod{419}$$

for all $T \in \mathbb{Z}_{419}$ yields

$$v_{419}(F(\lambda)) = v_{419}(\lambda - \Lambda), \qquad \frac{\lambda - \Lambda}{419^k} \equiv 233^{-1}u \pmod{419}.$$

Substitute these equalities into Lemma 3.1. The quadratic has valuation $1 + k$ on the right of (5). For even k, it is totally ramified. For odd k, it has even valuation and splits precisely when

$$\left(\frac{-10 \cdot 233^{-1}u}{419}\right) = \left(\frac{374u}{419}\right) = +1;$$

otherwise it is irreducible unramified. Since F is irreducible cubic, it has no rational zero: its critical repeated-root equality is impossible for integer λ. Finally $419^k \mid F(\lambda) \neq 0$, and the triangle inequality gives $|F(\lambda)| \leq C(1 + |\lambda|)^3$, proving (7). □

Remark 4.2. For example, $\lambda = 132$ gives $k = 1$ and an unramified quadratic, whereas $\lambda = 551 = 132 + 419$ gives $k = 1$ and a split quadratic. The deeper class $\lambda = 38261$ gives $k = 2$ and a totally ramified quadratic. The theorem needs no precomputed digits of Λ.

5 Prime-axis density: S_3 Chebotarev law

Theorem 5.1 (Uniform prime-density law)**.** *Fix any integer λ. Among primes $p \geq 7$, excluding the single prime* 419*, the natural densities of those for which $G_{p,\lambda}$ has exactly* 1*,* 2*, or* 4 *roots in $\mathbb{Q}_p$ are, respectively,*

$$\boxed{\delta_1 = \frac{1}{3}, \qquad \delta_2 = \frac{1}{2}, \qquad \delta_4 = \frac{1}{6}.} \tag{9}$$

The classification at each individual generic prime, not merely the densities, is independent of λ. In particular the average number of $\mathbb{Q}_p$-roots across primes tends to 2*.*

Proof. The reduction of H modulo 7 is $m^3 - 2m^2 - 2m + 2$ and has no root in $\mathbb{F}_7$, so the cubic H is irreducible over $\mathbb{Q}$. Its nonsquare discriminant $125700 = 100 \cdot 3 \cdot 419$ shows that its splitting-field Galois group is S_3. Frobenius cycle types in S_3 produce 0, 1, or 3 distinct roots of H modulo almost every prime with natural densities 2/6, 3/6, or 1/6, respectively, by the Chebotarev theorem [15]. Equation (3) adds exactly the simple root $m = 0$ and Theorem 2.1 lifts the roots. The weighted mean is $(1/3) + 2(1/2) + 4(1/6) = 2$. □

6 Parameter-axis density and precision law

Endow $\mathbb{Z}_{419}$ with normalized additive Haar measure μ; we may extend the quartic formally to all 419-adic parameters. The following distributions also hold as natural densities of integer parameters $0 \le \lambda < X$ when $X \to \infty$.

Theorem 6.1 (Exact parameter density)**.** *For the quadratic pair at the exceptional prime* 419*,*

$$\boxed{\begin{aligned} \mu(\textit{totally ramified}) &= \frac{419}{420}, \\ \mu(\textit{unramified irreducible}) &= \frac{1}{840}, \\ \mu(\textit{split into two linear factors}) &= \frac{1}{840}. \end{aligned}} \tag{10}$$

The critical equality $\lambda = \Lambda$ has measure zero and never occurs for integer λ. The mean number of distinct $\mathbb{Q}_{419}$-roots under Haar-random parameter is $841/420$*.*

Proof. Since translation preserves Haar measure, $\lambda - \Lambda$ is Haar-uniform. For every $j \ge 0$, $\mu(v_{419}(\lambda-\Lambda) = j) = 418/419^{j+1}$. The normal form (5) is totally ramified for j even, yielding the geometric sum

$$\sum_{a\ge 0} \frac{418}{419^{2a+1}} = \frac{419}{420}.$$

For j odd the valuation is even and the leading unit is uniformly random; precisely half are squares. Hence each of the split and unramified cases has measure

$$\frac{1}{2}\sum_{a\ge 0} \frac{418}{419^{2a+2}} = \frac{1}{840}.$$

At finite precision all decided branches are unions of congruence classes, while the single undecided congruence class has density tending to zero. This establishes the same natural densities for integer λ. The mean is $2 + 2(1/840) = 841/420$. □

Proposition 6.2 (Exact finite-precision residue counts)**.** *For each $N \ge 1$ there are exactly the following numbers of parameter classes modulo 419^N whose quadratic factorization is decided for* every *lift to $\mathbb{Z}_{419}$:*

$$S_N = U_N = 209 \sum_{j=0}^{\lfloor N/2\rfloor - 1} 419^{N-2j-2},$$
$$R_N = 418 \sum_{j=0}^{\lceil N/2\rceil - 1} 419^{N-2j-1}.$$

Here S_N, U_N, R_N denote split, unramified and totally ramified classes, respectively. Exactly one class, $\lambda \equiv \Lambda \pmod{419^N}$, remains undecided, and $S_N + U_N + R_N + 1 = 419^N$. For $N = 2$, the counts are $(209, 209, 175142, 1)$*.*

Proof. Among classes with exact $j = v_{419}(\lambda - \Lambda) < N$ there are $418 \cdot 419^{N-j-1}$ choices. If j is even all are ramified; if j is odd half split and half are unramified, by equidistribution of their leading units. Sum over the allowed parity classes. The only class with valuation $\ge N$ is $\lambda \equiv \Lambda \pmod{419^N}$. □

7 Prime-power congruence fibers and a resonance moment law

The number of distinct $\mathbb{Q}_{419}$-roots does not determine the number of roots modulo 419^n: singular residue classes can persist for several precision levels before disappearing. To capture this finer arithmetic, define

$$N_n(\lambda) = \#\{m \bmod 419^n : G_{419,\lambda}(m) \equiv 0 \pmod{419^n}\}, \qquad n \geq 1. \tag{11}$$

Write $P = 419$ for brevity, and retain $k = v_P(F(\lambda))$, $u = P^{-k}F(\lambda) \bmod P$ and $a = k + 1$ from Theorem 4.1.

Theorem 7.1 (Exact root counts at every prime-power precision)**.** *For every integer λ and $n \geq 1$, let $S_n(\lambda) = N_n(\lambda) - 2$. Then*

$$\boxed{S_n(\lambda) = \begin{cases} P^{\lfloor n/2 \rfloor}, & 1 \leq n \leq a, \\ 2P^{a/2}, & n > a,\ a \text{ even},\ \left(\frac{374u}{P}\right) = 1, \\ 0, & \text{otherwise.} \end{cases}} \tag{12}$$

In particular, $N_n(\lambda)$ is determined by finite integer arithmetic applied to $F(\lambda)$, for arbitrary n. Writing $N_0 = 1$, the unnormalized root-count series is rational and explicit:

$$\sum_{n \geq 0} N_n(\lambda) T^n = 1 + \frac{2T}{1-T} + \sum_{n=1}^{a} P^{\lfloor n/2 \rfloor} T^n + \mathbf{1}_{\text{split}} \frac{2P^{a/2} T^{a+1}}{1-T}, \tag{13}$$

where $\mathbf{1}_{\text{split}}$ is one exactly in the second case of (12)*.*

Proof. The reduction of $G_{419,\lambda}$ has two simple roots at $0, 20$ and one double root at -9. The first two contribute exactly two roots modulo each P^n by Hensel's lemma. In the disk $m \equiv c \equiv -9 \pmod P$, the coordinate $z = (m-c)\sqrt{U(m-c)/144}$ from Lemma 3.1 is an integral analytic isometry: its derivative is a unit congruent to one modulo P, and it induces a bijection on residue classes modulo P^n. Thus the singular contribution S_n equals the number of $z \bmod P^n$ with $z \equiv 0 \pmod P$ and $z^2 \equiv B \pmod{P^n}$, where $B = -10P(\lambda - \Lambda)$ and $v_P(B) = a$.

If $n \leq a$, this is $z^2 \equiv 0 \pmod{P^n}$ and has $P^{\lfloor n/2 \rfloor}$ solutions. If $n > a$, a solution requires $a = 2h$ even and $z = P^h w$. The reduced unit equation $w^2 \equiv B/P^{2h} \pmod{P^{n-2h}}$ has two Hensel roots exactly when its unit is a square modulo P; each gives P^h choices of z modulo P^n. The square criterion is precisely $\left(\frac{374u}{P}\right) = 1$ by Theorem 4.1. This proves the formula, and summing its eventually constant terms proves (13). □

Theorem 7.2 (Exact parameter moments and rare-resonance limit)**.** *Let λ be Haar-uniform in $\mathbb{Z}_P$. For every $n \geq 1$,*

$$\boxed{\mathbb{E}_\lambda N_n = 3, \qquad \operatorname{Var}_\lambda(N_n) = (P-1)(n-1) = 418(n-1).} \tag{14}$$

More precisely, with $S_n = N_n - 2$,

$$\Pr(S_n = P^{\lfloor n/2 \rfloor}) = P^{-(n-1)},$$
$$\Pr(S_n = 2P^h) = \frac{P-1}{2P^{2h}}, \quad 1 \leq h \leq \left\lfloor \frac{n-1}{2} \right\rfloor,$$

and the remaining probability is at $S_n = 0$. Moreover $N_n(\lambda)$ stabilizes almost surely as $n \to \infty$ to a finite N_∞, with

$$\Pr(N_\infty = 2 + 2P^h) = \frac{P-1}{2P^{2h}} \quad (h \geq 1), \qquad \Pr(N_\infty = 2) = \frac{839}{840}. \tag{15}$$

The limit has $\mathbb{E}N_\infty = 3$ but $\mathbb{E}N_\infty^2 = +\infty$, whereas the finite-n variances in (14) grow exactly linearly.

Proof. Translation by the critical parameter Λ preserves Haar measure, so $\Pr(v_P(\lambda-\Lambda) \geq n-1) = P^{-(n-1)}$. The first atom follows from (12). The second occurs when $v_P(\lambda - \Lambda) = 2h-1$ and the leading unit has the required quadratic character; its probability is $(P-1)/(2P^{2h})$. Since P is odd, the displayed positive values of S_n are distinct.

One may sum these atoms to obtain the first moment. Alternatively, for each $m \bmod P^n$ reducing to one of the three zeros $0, 20, -9$ modulo P, the congruence $G_{P,\lambda}(m) \equiv 0 \pmod{P^n}$ has Haar probability $P^{-(n-1)}$: the coefficient of λ is $1440P$ with $1440 \in \mathbb{Z}_P^\times$. Exactly $3P^{n-1}$ choices of m are eligible, whence $\mathbb{E}N_n = 3$. The second moment of the singular contribution is

$$\mathbb{E}S_n^2 = P^{2\lfloor n/2 \rfloor - (n-1)} + 2(P-1)\left\lfloor \frac{n-1}{2} \right\rfloor = 1 + (P-1)(n-1).$$

Since $\mathbb{E}S_n = 1$, the variance identity follows. Except on the singleton $\lambda = \Lambda$, the singular contribution eventually vanishes or stabilizes at $2P^h$, proving (15). Finally $\sum_{h\geq 1} 2P^h(P-1)/(2P^{2h}) = 1$, but $\sum_{h\geq 1}(2P^h)^2(P-1)/(2P^{2h}) = +\infty$. □

Remark 7.3 (Distinguishing local roots from congruence fibers)**.** The average number of *distinct* $\mathbb{Q}_{419}$-roots is $841/420$ by Theorem 6.1, not 3. An actual 419-adic root with derivative divisible by 419^h can account for 419^h residue solutions at every sufficiently high precision. The general theory and rationality of univariate local zeta/Poincaré functions are prior work; the present formulas are a CCR-specialized calculation, not a claim of inventing the generic method [13].

Corollary 7.4 (The exceptional prime 5 over parameters)**.** *For Haar-random $\lambda \in \mathbb{Z}_5$ the unique $\mathbb{Q}_5$-root α satisfies, for all $d \geq 0$,*

$$\Pr(v_5(\alpha) = 2 + d) = \frac{4}{5^{d+1}}, \quad \mathbb{E}\, v_5(\alpha) = \frac{9}{4}.$$

The other cubic is totally ramified for all parameters.

Proof. Use Theorem 2.1 and the geometric distribution of $v_5(\lambda - 115)$; the singleton parameter 115 has measure zero. □

8 Universal odd-column fold and p-adic source resonance

This section derives an extension to *all odd normalized CCR column orders $r \geq 5$*. Throughout, $\Pi_{p,r}(m)$ denotes the stationary normalized column supplied by the established higher-order CCR transport [3], not the d-orthogonal characteristic polynomial Q_r. The source parameters below r are fixed integers and the top source λ_r varies. The series identity used below belongs to classical Nörlund theory [9]; its role in the CCR deformation and the ensuing p-adic counts are the application here.

8.1 The all-order Stirling generating function

Let $S_r(m) = \Pi_{0,r}(m)$ be the formal $p = 0$ specialization and $S_0 = 1$. The original normalized polynomial transport yields

$$T_r S_r(m) = m S_{r-1}(m), \qquad (T_r f)(m) = m f(m+1) - (m-r) f(m). \tag{16}$$

Lemma 8.1 (Nörlund identity and universal double root)**.** *As an identity in* $\mathbb{Q}[m][[t]]$,

$$\sum_{r \geq 0} S_r(m) t^r = \left(\frac{t}{1-e^{-t}}\right)^m = e^{mt/2} \left(\frac{t}{2\sinh(t/2)}\right)^m. \tag{17}$$

For every $s \geq 1$*, the polynomial* S_{2s+1} *has an* exact double *root at zero:*

$$S_{2s+1}(m) = c_s m^2 + O(m^3), \qquad c_s = -\frac{B_{2s}}{4s(2s)!} \neq 0. \tag{18}$$

Here B_{2s} *is the classical Bernoulli number, and* $O(m^3)$ *is a polynomial divisible by* m^3*.*

Proof. Let $A(t) = t/(1-e^{-t})$. The generating form of (16) is $m(F(m+1,t) - F(m,t)) + t\partial_t F = mtF$. The proposed $F = A(t)^m$ satisfies this because $A - 1 + tA'/A = t$. Uniqueness follows since $T_r(m^d)$ is triangular with diagonal coefficient $r + d \neq 0$. The logarithm identity

$$\log A(t) = \frac{t}{2} - \sum_{j \geq 1} \frac{B_{2j}}{2j(2j)!} t^{2j}$$

shows that the coefficient of mt^{2s+1} vanishes and the coefficient of $m^2 t^{2s+1}$ is half the coefficient of t^{2s} in $\log A(t)$, giving (18). Classical Bernoulli numbers B_{2s} are nonzero. □

8.2 An all-odd-order arithmetic fold theorem

By stationary polynomiality, for each fixed r there exists an integer $D_r \geq 1$ which clears all rational coefficients of S_r and $(\Pi_{p,r} - S_r)/p$ in their independent formal variables. Define a *good prime at order* $r = 2s+1$ as an odd prime p not dividing rD_r and for which c_s in (18) is a p-adic unit. Only finitely many primes are excluded for each fixed r.

Theorem 8.2 (Universal odd-column CCR p-adic fold)**.** *Fix any odd* $r = 2s+1 \geq 5$*, any good prime* p*, and integer lower source parameters* $\lambda_4, \ldots, \lambda_{r-1}$*. The dependence of the stationary normalized polynomial on its top source is* exactly

$$\Pi_{p,r}^{(\lambda_r)}(m) = \Pi_{p,r}^{(0)}(m) - \frac{p}{r}\lambda_r. \tag{19}$$

There is a unique critical center $c \in p\mathbb{Z}_p$*, independent of* λ_r*, and a unique critical parameter* $\Lambda_r \in \mathbb{Z}_p$ *for which* $\Pi_{p,r}^{(\Lambda_r)}(c) = 0$*. An integral analytic isometric coordinate* z *on* $m \equiv 0 \pmod{p}$ *converts the local root equation exactly into*

$$\boxed{z^2 = K_r p(\lambda_r - \Lambda_r), \qquad K_r = \frac{1}{rc_s} = -\frac{4s(2s)!}{rB_{2s}} \in \mathbb{Z}_p^\times.} \tag{20}$$

If $k = v_p(\lambda_r - \Lambda_r) < \infty$*, the local degree-two factor is totally ramified and irreducible when* k *is even. When* k *is odd, it splits into two* $\mathbb{Q}_p$*-linear factors precisely when* $K_r(\lambda_r - \Lambda_r)/p^k$ *is a square unit modulo* p*, and otherwise is an unramified irreducible quadratic. At* $\lambda_r = \Lambda_r$*, the local factor has a repeated root. No nonattainment claim for integer parameters is made in this generality.*

Proof. In the order-r transport, the source λ_r first appears as the constant $(-1)^r p\lambda_r = -p\lambda_r$. Since T_r maps constants to r times themselves, subtracting the two transport equations gives (19) independently of the lower parameters. The good-prime condition ensures $\Pi_{p,r} \equiv S_r \pmod p$ coefficientwise, and Lemma 8.1 gives a nondegenerate double root at zero. Thus Hensel's lemma applied to the derivative of $\Pi^{(0)}_{p,r}$ yields the unique $c \in p\mathbb{Z}_p$. Because $\Pi^{(0)}_{p,r}(c) \equiv 0 \pmod p$ and p/r is a p times a unit, there is exactly one $\Lambda_r \in \mathbb{Z}_p$ canceling its critical value. Taylor expansion at c is

$$\Pi^{(\lambda_r)}_{p,r}(c+y) = -\frac{p}{r}(\lambda_r - \Lambda_r) + y^2 U(y), \qquad U(0) \equiv c_s \pmod p.$$

For $y \in p\mathbb{Z}_p$, the unit $U(y)/c_s$ has a unique analytic square root congruent to one modulo p, and $z = y\sqrt{U(y)/c_s}$ induces an isometric bijection of the disk. This proves (20). Odd-valuation critical values have quadratic Newton slope with denominator two, giving an Eisenstein/total-ramification conclusion; at even valuation the local equation lies within the integral disk and the p-adic square-unit criterion distinguishes split from unramified quadratic cases. The nonsingular complementary factor is separated by Hensel factorization. □

8.3 Universal local prime-power fiber and moment laws

For $n \geq 1$, count *only* the zero residue disk, rather than assuming all other residues are simple:

$$M_{p,r,n}(\lambda_r) = \#\{m \bmod p^n : m \equiv 0 \pmod p, \quad \Pi^{(\lambda_r)}_{p,r}(m) \equiv 0 \pmod{p^n}\}.$$

Theorem 8.3 (All-odd-order rare-resonance statistics)**.** *Under Theorem 8.2, write $a = 1 + v_p(\lambda_r - \Lambda_r)$ for a noncritical parameter. If a is even let ε be the Legendre symbol of the unit $K_r(\lambda_r - \Lambda_r)/p^{a-1}$. Then*

$$M_{p,r,n}(\lambda_r) = \begin{cases} p^{\lfloor n/2 \rfloor}, & n \leq a, \\ 2p^{a/2}, & n > a,\ a \text{ even},\ \varepsilon = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{21}$$

At the critical singleton, $M_{p,r,n} = p^{\lfloor n/2 \rfloor}$ for every n. For Haar-uniform $\lambda_r \in \mathbb{Z}_p$, with all lower parameters fixed, the exact local moments are

$$\boxed{\mathbb{E}M_{p,r,n} = 1, \qquad \mathrm{Var}(M_{p,r,n}) = (p-1)(n-1).} \tag{22}$$

The almost-sure limiting count M_∞ has mass $(p-1)/(2p^{2h})$ at $2p^h$ for every $h \geq 1$, and remaining mass $1 - 1/[2(p+1)]$ at zero; it has mean one and infinite second moment.

Proof. The local coordinate of Theorem 8.2 bijects every finite residue disk, so the count equals the solutions of $z^2 \equiv K_r p(\lambda_r - \Lambda_r) \pmod{p^n}$ with $z \equiv 0 \pmod p$. The same valuation and Hensel count as in Theorem 7.1 gives (21). Haar translation makes $\lambda_r - \Lambda_r$ uniform. The positive atoms have probabilities $p^{-(n-1)}$ at $p^{\lfloor n/2 \rfloor}$ and $(p-1)/(2p^{2h})$ at $2p^h$, for $1 \leq h \leq \lfloor (n-1)/2 \rfloor$. Their first moment is one; their second moment is $1 + (p-1)(n-1)$. The limit follows by summing the stable split atoms; its second moment diverges by constant contributions over infinitely many values of h. □

8.4 The entire fifth column away from $p = 73$

For $r = 5$, the sharp denominator $D_5 = 11520$ is an earlier fixed-order CCR result and $c_2 = 1/5760$. Thus *every prime $p \geq 7$ is good for the local fifth-column fold*, with $K_5 = 1152$. The inherited Stirling polynomial is

$$S_5(m) = \frac{m^2(m-1)(3m^2 - 7m - 2)}{11520}. \tag{23}$$

Moreover, writing $N_5 = 11520\Pi_{p,5}$, direct use of the original transport yields

$$\begin{aligned}
[m^5]N_5 &= 3(p+1)^5, \\
[m^4]N_5 &= 10(p+1)^4(4p-1), \\
[m^3]N_5 &= -5(p+1)^3(107p^2 + 74p - 1), \\
[m^2]N_5 &= 2(p+1)^2(280p^3 + 705p^2 + 106p + 1), \\
[m]N_5 &= 4p(p+1)(360\lambda_4 + 1183p^3 + 1157p^2 + 629p + 295), \\
[1]N_5 &= -96p(66\lambda_4 p + 36\lambda_4 + 24\lambda_5 + 85p^4 + 138p^3 + 76p^2 + 30p + 13).
\end{aligned}$$

Consequently

$$\Lambda_5(p, \lambda_4) \equiv -\frac{36\lambda_4 + 13}{24} \pmod{p}. \tag{24}$$

Corollary 8.4 (Global fifth-column congruence moments)**.** *Fix $p \geq 7$, $p \neq 73$, and integer λ_4. Let $N_{p,n}(\lambda_5)$ count every root of $\Pi_{p,5}$ modulo p^n. With $\chi_p = \left(\frac{73}{p}\right) \in \{-1, 1\}$, the complete count is*

$$N_{p,n}(\lambda_5) = \underbrace{(2 + \chi_p)}_{\text{simple residual roots}} + M_{p,5,n}(\lambda_5). \tag{25}$$

For Haar-uniform $\lambda_5 \in \mathbb{Z}_p$,

$$\boxed{\mathbb{E}N_{p,n} = 3 + \chi_p, \qquad \mathrm{Var}(N_{p,n}) = (p-1)(n-1).} \tag{26}$$

Proof. After excluding $p = 73$, the reduction (23) has exactly one double zero at 0 and all other zeros are simple. The simple zero at $m = 1$ always lifts. The quadratic $3m^2 - 7m - 2$ has discriminant 73, hence has exactly $1 + \chi_p \in \{0, 2\}$ simple $\mathbb{F}_p$-roots. These yield $2 + \chi_p$ lifts at all precisions. Add the singular contribution of Theorem 8.3. □

Remark 8.5 (Boundary and novelty status)**.** The exceptional prime 73 has a second repeated residue; its complete global classification is proved separately in Section 9. For $r \geq 7$ the all-odd-order theorem is intentionally local to the zero residue disk: other singular residues are not assumed away. The Nörlund identity [9], the standard analytic square coordinate and the moment-summation method are classical; the qualified research advance is this exact CCR-specific source deformation for every odd order.

9 The exceptional fifth-column prime 73: two coupled folds

The discriminant of $3m^2 - 7m - 2$ is 73. Thus the prime excluded from Corollary 8.4 is not merely a bad denominator: it creates a *second* nondegenerate critical residue. Throughout this section put $p = 73$, $L = \lambda_4$, $T = \lambda_5$, and denote the integral fifth-column numerator by $N_{L,T}(m) = 11520\Pi_{73,5}^{(L,T)}(m)$. The coefficient formula above shows that the dependence on T is exactly $-2304 \cdot 73T$ and

$$N_{L,T}(m) \equiv 9m^2(m-1)(m-62)^2 \pmod{73}. \tag{27}$$

The residue $m = 1$ is simple; the two repeated residues 0 and 62 have nonzero quadratic Taylor coefficients 2 and 24, respectively, modulo 73.

Theorem 9.1 (Two-fold normal form and critical-value collision)**.** *For every $L \in \mathbb{Z}_{73}$ there are unique critical centers $c_0(L) \equiv 0$ and $c_{62}(L) \equiv 62 \pmod{73}$, and two critical source values $\Lambda_0(L), \Lambda_{62}(L) \in \mathbb{Z}_{73}$. The two residue disks admit integral analytic isometric coordinates in which their equations are, respectively,*

$$z_0^2 = 1152 \cdot 73\,[T - \Lambda_0(L)], \qquad z_{62}^2 = 96 \cdot 73\,[T - \Lambda_{62}(L)]. \tag{28}$$

Their critical parameters have first digits

$$\Lambda_0(L) \equiv 39 + 35L, \qquad \Lambda_{62}(L) \equiv 37 + 19L \pmod{73}. \tag{29}$$

In particular, their reductions coincide precisely when $L \equiv 9 \pmod{73}$. There is exactly one $L_* \in \mathbb{Z}_{73}$ for which $\Lambda_0(L_*) = \Lambda_{62}(L_*)$; its first four base-73 digits are

$$L_* \equiv 9 + 11 \cdot 73 + 30 \cdot 73^2 + 35 \cdot 73^3 \pmod{73^4}. \tag{30}$$

At the pair $(L_*, \Lambda_0(L_*))$, the quintic has two distinct double roots and one simple root in $\mathbb{Q}_{73}$.

Proof. Apply Hensel's lemma to $\partial_m N_{L,T}$ in the two disks: its reduction vanishes simply at both residue centers by the nonzero second derivatives. The derivative does not depend on T, hence has unique 73-adic centers $c_i(L)$ independent of T. Put $\Lambda_i(L) = N_{L,0}(c_i)/(2304 \cdot 73)$; integrality follows from (27). In each disk the exact Taylor expansion is

$$N_{L,T}(c_i + y) = -2304 \cdot 73(T - \Lambda_i) + y^2 U_i(y),$$

where $U_0(y) \equiv 2$ and $U_{62}(y) \equiv 24 \pmod{73}$. Therefore $z_i = y\sqrt{U_i(y)/u_i}$, $u_0 = 2$, $u_{62} = 24$, is integral analytic and induces an isometric bijection on every congruence quotient of its residue disk. To see the global assertion, write $s_i(y) = \sqrt{U_i(y)/u_i}$. On $73\mathbb{Z}_{73}$, $s_i(y) \equiv 1 \pmod{73}$ and $s_i(y) - s_i(y') \in (y - y')\mathbb{Z}_{73}$. Consequently $z_i(y) - z_i(y') = (y - y')(s_i(y) + y'\,(s_i(y) - s_i(y'))/(y - y'))$, whose second factor is a 73-adic unit. The inverse is obtained by successive unique Hensel lifts. The equations become (28) because $2304/2 = 1152$ and $2304/24 = 96$.

Since $c_i \equiv i \pmod{73}$ and $N'_{L,0}(c_i) = 0$, the first digit of Λ_i is obtained by evaluating $N_{L,0}(i)/(2304 \cdot 73)$ modulo 73. The displayed exact coefficient formula yields (29). Differentiating through the critical center,

$$\Lambda_i'(L) = \frac{\partial_L N_{L,0}(c_i(L))}{2304 \cdot 73},$$

so $\Lambda_0' - \Lambda_{62}' \equiv 35 - 19 = 16 \pmod{73}$ is a unit for all L. Analytic Hensel lifting of $\Lambda_0(L) - \Lambda_{62}(L)$ gives a unique zero in the residue class $L \equiv 9 \pmod{73}$. Successive exact derivative-center lifts give (30). The residue disks remain distinct, so at the simultaneous critical parameter there are two double roots, plus the simple lift of 1. □

For $i \in \{0, 62\}$ and $n \geq 1$, let $M_{i,n}(T)$ count roots modulo 73^n in residue class i of $N_{L,T}$. These are the separate quadratic-fiber counts from Theorem 8.3, with constants $K_0 = 1152$, $K_{62} = 96$ and critical sources Λ_i. The ratio $K_0/K_{62} = 12$ is a square in $\mathbb{Z}_{73}^\times$: indeed $31^2 \equiv 12 \pmod{73}$.

Theorem 9.2 (Exact 73-adic global count and covariance law)**.** *For fixed $L \in \mathbb{Z}_{73}$ and any $T \in \mathbb{Z}_{73}$, the entire quintic root count at every precision $n \geq 1$ is*

$$N_n(L,T) = 1 + M_{0,n}(T) + M_{62,n}(T). \tag{31}$$

For Haar-uniform T and fixed L, $\mathbb{E}_T N_n = 3$ for every $n \geq 1$. Put $d = v_{73}(\Lambda_{62}(L) - \Lambda_0(L)) \in \{0, 1, 2, \ldots, \infty\}$, with $v_{73}(0) = \infty$. At $n = 1$, $N_1 = 3$ deterministically. For $n \geq 2$, the exact mixed moment of the two singular contributions is

$$\mathbb{E}_T[M_{0,n}M_{62,n}] = \begin{cases} 0, & d = 0, \\ 1 + 72(n-1), & d \geq 1, \quad n \leq d+1, \\ 72d, & 1 \leq d < \infty, \quad n \geq d+2. \end{cases} \tag{32}$$

Consequently, with $\mathrm{Var}(N_1) = 0$*,*

$$\boxed{\mathrm{Var}_T(N_n) = \begin{cases} 144(n-1) - 2, & d = 0, \\ 288(n-1), & d \geq 1, \quad n \leq d+1, \\ 144(n-1) + 144d - 2, & 1 \leq d < \infty, \quad n \geq d+2. \end{cases}} \tag{33}$$

For $d = \infty$ the second line applies at all $n \geq 2$, reflecting perfectly correlated fold counts.

Proof. The factorization (27) and Hensel's lemma give one root in the simple disk at every 73^n, establishing (31). Each quadratic disk has mean one and variance $72(n-1)$ by Theorem 8.3.

For $n \geq 2$ write the local coordinate $z_i = 73a_i$ with $a_i \pmod{73^{n-1}}$. Equation (28) becomes

$$T \equiv \Lambda_i + \frac{73}{K_i} a_i^2 \pmod{73^{n-1}}.$$

Let $\delta = \Lambda_{62} - \Lambda_0$. Expanding the product of the finite solution counts and averaging over uniform $T \pmod{73^{n-1}}$ gives

$$\mathbb{E}[M_{0,n}M_{62,n}] = 73^{-(n-1)} \# \left\{ (a,b) \pmod{73^{n-1}} : 73\Big(\frac{a^2}{1152} - \frac{b^2}{96}\Big) \equiv \delta \pmod{73^{n-1}} \right\}.$$

If $d = 0$, this set is empty. For $d \geq 1$, put $t = n - 2$ and divide by 73. Each variable has one additional free base-73 digit, so the last display equals

$$73^{1-t} \#\{(a,b) \pmod{73^t} : a^2 - 12b^2 \equiv 1152\delta/73 \pmod{73^t}\}.$$

Choose $s \in \mathbb{Z}_{73}^\times$ with $s^2 = 12$. The invertible linear transformation $(a,b) \mapsto (u,v) = (a - sb, a + sb)$ reduces the count to $uv \equiv 1152\delta/73 \pmod{73^t}$. If $0 \leq d - 1 < t$, each valuation $v_{73}(u) = 0, \ldots, d-1$ supplies $72 \cdot 73^{t-1}$ pairs, giving $72d$ after normalization. If $d - 1 \geq t$ (including $d = \infty$), the zero-product congruence has $73^t + t \cdot 72 \cdot 73^{t-1}$ pairs, giving $1 + 72(n-1)$. The case $t = 0$ is the single trivial residue class, with the same count by direct interpretation. This proves (32). Since each local mean is one, the variance of the sum of the two folds is $2 \cdot 72(n-1) + 2(\mathbb{E}[M_{0,n}M_{62,n}] - 1)$, proving (33). □

Corollary 9.3 (Attainability of complete local splitting)**.** *For every fixed ordinary integer L, there exists an ordinary integer T for which $N_{L,T}$ has five distinct $\mathbb{Q}_{73}$ roots if and only if*

$$L \equiv 9 \pmod{73}. \tag{34}$$

In particular, $L = 9$, $T = 135$ supplies a witness with

$$(N_1, N_2, N_3, N_4) = (3, 147, 293, 293),$$

and both quadratic factors split over $\mathbb{Q}_{73}$.

Proof. If $L \not\equiv 9 \pmod{73}$, the critical parameter residues are distinct. Splitting of a fold requires T to lie in its critical residue class modulo 73, so both cannot split. If $L \equiv 9$, put $d = v_{73}(\Lambda_{62} - \Lambda_0) \geq 1$. For $d \geq 2$, take T with $v_{73}(T - \Lambda_0) = 1$ and with the square leading unit specified by K_0; then the second fold has the same valuation and compatible square character, because $K_0/K_{62} = 12$ is a 73-adic square. Such T is available as an ordinary integer by choosing its residue modulo 73^2. For $d = 1$, write $T = \Lambda_0 + 73u$ and $\delta = 73\Delta$ with Δ a unit. Let χ be the quadratic character modulo 73 extended by $\chi(0) = 0$. Both split if $\chi(K_0 u) = \chi(K_{62}(u - \Delta)) = 1$. Since $\chi(K_0) = \chi(K_{62})$ and $\chi(-1) = 1$, a standard quadratic-character correlation gives exactly

$$\frac{73 - 3 - 2\chi(K_0\Delta)}{4} \in \{17, 18\}$$

admissible residues $u \pmod{73}$, hence an integer T exists. Direct evaluation at $L = 9$ yields $\Lambda_0 \equiv 5026$, $\Lambda_{62} \equiv 1887 \pmod{73^2}$; $T = 135$ is one of the admissible residues. The displayed modular counts are independently verified by direct integer-polynomial Hensel lifting, and the split local normal forms prove the assertion about five $\mathbb{Q}_{73}$-roots. □

Remark 9.4 (Scope of the collision phenomenon)**.** The parameter L_* is a 73-adic analytic critical-value coincidence, not a claim that L_* is an ordinary integer. The local analytic folds and character-sum/product-congruence counts are standard methods; the explicit two-fold coupling and the exact finite-precision covariance stratification belong to the concrete CCR fifth-column family. They do not establish global all-order physical DPDA realization or a new generic p-adic Morse theorem.

10 Joint source equidistribution and the exact five-root density at 73

The preceding section fixed the lower stationary source $L = \lambda_4$ and varied $T = \lambda_5$. We now vary both L, T independently according to normalized Haar measure on $\mathbb{Z}_{73}^2$. This change reveals a sharp distinction between *conditional* dependence and *unconditional* independence of the two singular residue disks.

Theorem 10.1 (Two-source critical-value isometry)**.** *Retain the two critical parameter functions $A(L) = \Lambda_0(L)$ and $B(L) = \Lambda_{62}(L)$ of Section 9. The difference $D(L) = B(L) - A(L)$ is an isometric bijection $\mathbb{Z}_{73} \to \mathbb{Z}_{73}$. In fact, for every $L \neq L'$ in $\mathbb{Z}_{73}$,*

$$\frac{D(L) - D(L')}{L - L'} \equiv 19 - 35 = -16 \pmod{73}. \tag{35}$$

Consequently the map

$$\Phi : (L, T) \longmapsto \big(T - A(L),\ T - B(L)\big)$$

is a measure-preserving bijection of $\mathbb{Z}_{73}^2$. Thus its two output coordinates are independent Haar-uniform 73-adic integers when (L, T) is Haar-uniform.

Proof. The established quintic numerator $N_{L,T}(m)$ is affine in L and T, with L- and T-coefficients divisible by 73; moreover the T-coefficient is the constant $-2304 \cdot 73$. Let $c_i(L)$ be the stationary derivative-zero center for $i = 0, 62$. Since $\partial_m^2 N_{L,0}(c_i(L))$ is a unit modulo 73, implicit Hensel lifting makes c_i an integral analytic function of L. As $\partial_L \partial_m N_{L,0}$ is divisible by 73,

$$c_i(L) - c_i(L') \in 73(L - L')\mathbb{Z}_{73}.$$

By definition $A(L) = N_{L,0}(c_0(L))/(2304 \cdot 73)$ and likewise $B(L) = N_{L,0}(c_{62}(L))/(2304 \cdot 73)$. The vanishing of the m-derivative at each center, combined with the preceding Lipschitz estimate, shows that the critical-value difference quotients reduce modulo 73 to the explicit L-coefficient at the corresponding residue center. The exact coefficient formula in the previous section gives

$$\frac{A(L) - A(L')}{L - L'} \equiv 35, \qquad \frac{B(L) - B(L')}{L - L'} \equiv 19 \pmod{73}.$$

This proves (35). It also proves that D is an isometry. Since $D(L) \equiv -2 - 16L \pmod{73}$ is a permutation, for each target $t \in \mathbb{Z}_{73}$ and every solution L_j of $D(L_j) \equiv t \pmod{73^j}$ there is exactly one next digit $h \pmod{73}$ such that $D(L_j + h73^j) \equiv t \pmod{73^{j+1}}$: the difference quotient above makes the digit coefficient -16, a unit. Induction proves that D is bijective modulo every 73^j and preserves Haar measure.

Now put $(U, V) = \Phi(L, T)$. The inverse is explicit:

$$L = D^{-1}(U - V), \qquad T = U + A(L).$$

Both the forward and inverse maps preserve congruence classes modulo every 73^j and give bijections of their finite residue grids; therefore Φ preserves product Haar measure. In particular U, V are independent Haar variables. □

Theorem 10.2 (Complete joint law for the fifth column)**.** *For Haar-independent $L, T \in \mathbb{Z}_{73}$ and every $n \geq 1$, the two singular-disk solution counts $M_{0,n}, M_{62,n}$ are independent and identically distributed. The complete fifth-column congruence count $N_n = 1 + M_{0,n} + M_{62,n}$ obeys*

$$\mathbb{E}_{L,T} N_n = 3, \qquad \mathrm{Var}_{L,T}(N_n) = 144(n-1). \tag{36}$$

More precisely, write $h = \lfloor (n-1)/2 \rfloor$, $q_n = 73^{-(n-1)}$, $a_n = 73^{\lfloor n/2 \rfloor}$, and $w_j = 36/73^{2j}$. The exact probability generating polynomial is

$$P_n(X) = \left(1 - q_n - \sum_{j=1}^{h} w_j\right) + q_n X^{a_n} + \sum_{j=1}^{h} w_j X^{2 \cdot 73^j}, \tag{37}$$

$$\mathbb{E}_{L,T}[X^{N_n}] = X P_n(X)^2.$$

Let R be the number of distinct roots in $\mathbb{Q}_{73}$ of the fifth-column quintic. Then

$$\Pr(R = 1) = \frac{21609}{21904}, \quad \Pr(R = 3) = \frac{294}{21904}, \quad \Pr(R = 5) = \boxed{\frac{1}{21904}}, \quad \mathbb{E}R = \frac{38}{37}. \tag{38}$$

These probabilities are also the natural densities of ordinary integer source pairs (L, T) in expanding integer squares. The almost-surely stabilized congruence count has expectation 3 but infinite variance.

Proof. The local formulas of Section 8 depend on (L, T) only through the appropriate difference $U = T - A(L)$ or $V = T - B(L)$. Theorem 10.1 makes these independent Haar inputs. Each local count has mean 1, variance $72(n-1)$ and probability polynomial $P_n(X)$, as proved previously; their sum gives (36) and (37). Each quadratic splits over $\mathbb{Q}_{73}$ with probability $s = 1/(2 \cdot 74) = 1/148$, independently. The simple residue root $m \equiv 1$ always survives; therefore $R = 1 + 2I_0 + 2I_{62}$, where I_0, I_{62} are independent Bernoulli variables with parameter s, outside a Haar-null repeated-root locus. This yields (38). The sets specifying splitting are unions of 73-adic cylinders up to null critical singletons. Approximating by finite cylinders proves equality of Haar probabilities with the natural densities among integer pairs. The limit-moment statements follow from the independent heavy-tailed local limits already proved for one fold. □

Proposition 10.3 (Finite two-digit certificate)**.** *Modulo* 73^2*, the critical gap map* $L \mapsto D(L)$ *permutes all* 5329 *residue classes. Among the* $73^4 = 28\,398\,241$ *source pairs* (L, T) *modulo* 73^2*, exactly*

$$\left(\frac{73-1}{2}\right)^2 = \boxed{1296}$$

classes certify that both folds split at their first possible resonance depth $v_{73}(T - \Lambda_i) = 1$*. For the* 73 *values of* $L \pmod{73^2}$ *with* $L \equiv 9 \pmod{73}$*, precisely* 36 *admit* 18 *such* T*,* 36 *admit* 17*, and one admits* 36*. Moreover the full* N_2 *distribution on* $(L, T) \pmod{73}$ *is*

$$\#\{N_2 = 1\} = 5184, \qquad \#\{N_2 = 74\} = 144, \qquad \#\{N_2 = 147\} = 1.$$

Proof. By Theorem 10.1, (U, V) ranges bijectively over the pair of residue classes modulo 73^2. A first-depth split in each coordinate requires a nonzero square leading unit, for which there are $(73-1)/2 = 36$ possibilities. This gives 1296. The finer $36, 36, 1$ census follows from the two explicit critical functions and a quadratic character sum (or direct enumeration of all 73 coincident lower-source residues). At precision $n = 2$, each disk contributes 73 solutions iff its critical source residue matches $T \pmod{73}$, otherwise none. The bijection $\Phi \pmod{73}$ therefore gives the stated binomial census. □

Remark 10.4 (Conditional and unconditional source correlations)**.** The independent global law does not negate the sharply nonzero *conditional* covariance of Theorem 9.2: fixing L creates dependence, which cancels exactly on averaging through the measure-preserving lower-source coordinate. The isometry/Haar principle is standard in p-adic dynamics; the specific CCR two-critical-value map, explicit simultaneous-splitting density, and joint source distribution are the concrete arithmetic statements established here. The conclusions do not include a generic isometry theorem or a theorem on physical source realizability.

11 Fixed integer sources: finite resonance support across primes

The preceding root-density results distinguish two different sampling schemes: variation of a stationary source in $\mathbb{Z}_p$ with p fixed, versus variation of the prime while the sources are fixed *ordinary integers.* The latter exhibits a rigid arithmetic obstruction not apparent from the Haar probability of splitting at any one prime.

Theorem 11.1 (Fixed-source prime support and exact two-digit stability)**.** *Fix* $L = \lambda_4, T = \lambda_5 \in \mathbb{Z}$ *and put*

$$A(L, T) = 24T + 36L + 13 = 12(2T + 3L + 1) + 1 \neq 0.$$

For every rational prime $p \geq 7$ *satisfying* $p \neq 73$ *and* $p \nmid A(L, T)$*, the integer fifth-column numerator* $N_{p,L,T}(m) = 11520\Pi_{p,5}^{(L,T)}(m)$ *has exactly*

$$b_p = 2 + \left(\frac{73}{p}\right) \in \{1, 3\}$$

distinct roots in $\mathbb{Q}_p$*. More precisely, letting* $C_{p,n}$ *count solutions of* $N_{p,L,T}(m) \equiv 0 \pmod{p^n}$*,*

$$C_{p,1} = 3 + \left(\frac{73}{p}\right), \qquad C_{p,n} = 2 + \left(\frac{73}{p}\right) \quad (n \geq 2).$$

All surviving roots arise from simple residues, while the double residue $m \equiv 0 \pmod{p}$ *yields a totally ramified irreducible quadratic over* $\mathbb{Q}_p$ *and disappears already modulo* p^2*.*

Proof. The exact coefficient formula of the previous section gives

$$N_{p,L,T}(m) \equiv m^2(m-1)(3m^2-7m-2) \pmod{p}.$$

For $p \geq 7$, $p \neq 73$, the nonzero roots of the right side are simple: $m = 1$, plus two roots of its quadratic factor exactly when $\left(\frac{73}{p}\right) = 1$. Hensel's lemma lifts each such root uniquely through all p^n and to $\mathbb{Q}_p$.

The exact constant and linear coefficients satisfy

$$N_{p,L,T}(0) \equiv -96p(24T + 36L + 13) \pmod{p^2}, \qquad [m]N_{p,L,T} \equiv 0 \pmod{p}.$$

Consequently, for every $z \in \mathbb{Z}_p$,

$$N_{p,L,T}(pz) \equiv -96pA(L,T) \pmod{p^2}.$$

The right side is nonzero modulo p^2 when $p \nmid A$, as $-96 \in \mathbb{Z}_p^\times$. Thus *no* zero-disk solution survives even to p^2. Its local quadratic factor has Newton slope $-1/2$ (nonzero constant valuation one, unit quadratic coefficient), hence is irreducible and totally ramified. This proves the exact congruence and field-root counts. Finally $A(L,T) \equiv 1 \pmod{12}$, so no integral source pair makes $A = 0$. □

Corollary 11.2 (Source-independent prime-axis densities)**.** *For every fixed integer pair (L,T) the fifth-column quintic has exactly one $\mathbb{Q}_p$-root at a natural density $1/2$ of rational primes, and exactly three at density $1/2$; primes exhibiting five roots are contained in the* finite *exceptional set*

$$\{73\} \cup \{p \geq 7 : \ p \mid (24T + 36L + 13)\}$$

(aside from the excluded small primes). The average local-root count over primes tends to 2. Moreover, for any $B \geq 0$, all integer sources $|L|, |T| \leq B$ satisfy Theorem 11.1 simultaneously for every $p > \max\{73, 60B + 13\}$.

Proof. Since $73 \equiv 1 \pmod 4$, quadratic reciprocity gives $\left(\frac{73}{p}\right) = \left(\frac{p}{73}\right)$. Exactly 36 of the 72 nonzero residues modulo 73 are squares; Dirichlet's theorem on primes in arithmetic progressions assigns density $1/72$ to each residue class, so the two root counts each have density $1/2$. Only finitely many primes divide the nonzero integer A. Finally $|A| \leq 24|T| + 36|L| + 13 \leq 60B + 13$, which yields the uniform threshold. □

Remark 11.3 (Comparison with source-Haar statistics)**.** The density-1/2 prime dichotomy concerns *fixed integer* sources; it is not the source-Haar distribution of Theorem 8.3, and it does not override the exceptional prime 73 two-fold model. Standard Dirichlet and Hensel methods are used; the new CCR-specific content is the exact integer resonance obstruction $A(L,T)$ and its two-digit criterion.

12 All odd orders: a Stirling source obstruction and finite prime support

The fifth-column obstruction of the previous section is not isolated. In this section we derive a closed source-obstruction formula *for every odd column*, using the pre-existing normalized transport and the classical Nörlund generating function. The odd-index double-zero property itself has appeared in the Stirling-polynomial literature [4]; the CCR source obstruction and its prime-support application are the subject here.

Write $\left\{{n \atop k}\right\}$ for a Stirling number of the second kind. Throughout, $r \geq 3$, $\lambda_3 = 1$, and all λ_a are independent stationary source parameters unless explicitly specialized to integers. Denote by $S_r(m) = \Pi_{0,r}(m)$ the Stirling specialization and set $E_r(p, m, \boldsymbol{\lambda}) = (\Pi_{p,r}(m) - S_r(m))/p$ in the rational coefficient ring supplied by the original polynomial transport.

Theorem 12.1 (Exact first-order source obstruction)**.** *For every $r \geq 3$ one has*

$$A_r(\boldsymbol{\lambda}) := \sum_{a=3}^{r} (a-1)! \left\{ {r-1 \atop a-1} \right\} \lambda_a - 1, \qquad E_r(0,0,\boldsymbol{\lambda}) = \frac{(-1)^{r-2}}{r!} A_r(\boldsymbol{\lambda}). \tag{39}$$

For integer stationary sources, A_r is always odd. For every odd $r \geq 5$, the stronger arithmetic rigidity holds:

$$A_r = 2^{r-1} - 3 + \sum_{a=4}^{r} (a-1)! \left\{ {r-1 \atop a-1} \right\} \lambda_a \equiv 1 \pmod 6. \tag{40}$$

In particular $A_r(\boldsymbol{\lambda}) \neq 0$ for every *integer vector $\boldsymbol{\lambda}$.*

Proof. Apply the established stationary transport (1) as a polynomial identity. Differentiate formally at $p = 0$, put $m = 0$, use $S_j(0) = 0$ for $j \geq 1$, and note $T_j f(0) = j f(0)$. This gives

$$j E_j(0,0) = -S_{j-2}(-1) + \sum_{a=3}^{j} (-1)^a \lambda_a S_{j-a}(1-a). \tag{41}$$

The established Nörlund identity $\sum_{q \geq 0} S_q(m) t^q = (t/(1-e^{-t}))^m$ yields

$$S_{j-2}(-1) = \frac{(-1)^{j-2}}{(j-1)!}, \qquad S_{j-a}(1-a) = \frac{(-1)^{j+a-2}(a-1)!}{(j-1)!} \left\{ {j-1 \atop a-1} \right\}.$$

The second formula follows by expanding $(1-e^{-t})^{a-1}/t^{a-1}$ and using the standard finite-difference identity defining Stirling numbers. Substituting these coefficients into (41) proves (39). All $(a-1)!$ are even for $a \geq 3$, so the sum minus one is odd. Since $2!\{{r-1 \atop 2}\} = 2(2^{r-2}-1)$ and $(a-1)!$ is divisible by 6 for $a \geq 4$, equation (40) follows when $r-1$ is even and at least four. □

Theorem 12.2 (Exact CCR source-obstruction lattice and coherent integral witness)**.** *For every $r \geq 4$ put $n = r-1$ and define the positive integer*

$$d_r = \gcd_{3 \leq j \leq n} \left(j! \left\{ {n \atop j} \right\} \right).$$

With $\lambda_3 = 1$ and all subsequent stationary sources ranging freely over $\mathbb{Z}$, the exact image *of the first-order source obstruction is*

$$\{ A_r(1, \lambda_4, \ldots, \lambda_r) : \lambda_a \in \mathbb{Z} \} = (-1)^{r-1} + d_r \mathbb{Z}. \tag{42}$$

There is a single coherent choice of integer sources for every order,

$$\lambda_a^* = (-1)^{a-1} \quad (a \geq 3), \qquad A_r(\boldsymbol{\lambda}^*) = (-1)^{r-1} \quad (r \geq 3). \tag{43}$$

For every odd $r \geq 5$, the image in (42) *is $1 + d_r \mathbb{Z}$, and $12 \mid d_r$. In the first three odd cases,*

$$d_5 = 12, \qquad d_7 = 60, \qquad d_9 = 252.$$

Thus (40) *may be strengthened to the exact lattice image, not just a necessary congruence.*

Proof. The established Stirling finite-difference identity [8, Eq. 26.8.36] is $\sum_{j=0}^{n} (-1)^j j! \{{n \atop j}\} = (-1)^n$. At $n \geq 2$ the $j = 0$ term vanishes and the $j = 1$ term equals -1. Substituting

$\lambda^*_{j+1} = (-1)^j$ into (39) therefore yields $A_r(\boldsymbol{\lambda}^*) = (-1)^n$ exactly. Subtracting the canonical-source equality from the general source gives

$$A_r(\boldsymbol{\lambda}) - (-1)^n = \sum_{j=3}^{n} j! \left\{ {n \atop j} \right\} (\lambda_{j+1} - (-1)^j).$$

Every right-hand term is divisible by d_r. Conversely, Bézout's identity shows that arbitrary integer values of the freely chosen source differences generate *every* element of $d_r\mathbb{Z}$. This proves the exact image rather than only a containment. For odd $r \geq 5$, $n \geq 4$ is even. The terms $j!$ for $j \geq 4$ are multiples of 24. The remaining generator satisfies

$$3! \left\{ {n \atop 3} \right\} = 3^n - 3 \cdot 2^n + 3 \equiv 0 \pmod{12},$$

since $3^n \equiv 9 \pmod{12}$ and $3 \cdot 2^n \equiv 0 \pmod{12}$. Hence $12 \mid d_r$. Exact Stirling arithmetic gives the three displayed small-order gcd values. □

Proposition 12.3 (Closed form of the odd-order source-lattice index)**.** *Let $r \geq 5$ be odd, and let d_r be the index in Theorem 12.2. Then*

$$\boxed{d_r = 4 \prod_{\substack{\ell \geq 3 \text{ prime} \\ \ell - 1 \mid r-3}} \ell^{1 + \mathbf{1}_{\ell \mid r-3}}.} \tag{44}$$

Equivalently, for every odd prime ℓ,

$$\ell \mid d_r \iff \ell - 1 \mid r - 3, \qquad \ell^2 \mid d_r \iff \ell(\ell-1) \mid r - 3, \qquad \ell^3 \nmid d_r,$$

and $v_2(d_r) = 2$. In particular, the exact source image $1 + d_r\mathbb{Z}$ is determined solely by the elementary prime factorizations of the divisors of $r-3$; its first values are $12, 60, 252, 60, 132, 16380$ at $r = 5, 7, 9, 11, 13, 15$.

Proof. Write $n = r - 1 \geq 4$ (even), set $a = 2^{n-1} - 1$, and introduce the integer polynomial

$$P_n(X) = X^n - X - aX(X-1).$$

By $X^n = \sum_{j=0}^{n} j! \left\{ {n \atop j} \right\} \binom{X}{j}$ and $2! \left\{ {n \atop 2} \right\} = 2^n - 2 = 2a$, its binomial-basis expansion is

$$P_n(X) = \sum_{j=3}^{n} j! \left\{ {n \atop j} \right\} \binom{X}{j}.$$

Integral forward differences and binomial inversion therefore identify d_r with the fixed divisor $\gcd\{P_n(x) : x \in \mathbb{Z}\}$: taking differences recovers every binomial coefficient, and conversely every integer polynomial value is an integral combination of them.

Let $\ell \geq 5$ be prime. The condition $\ell \mid d_r$ means that P_n vanishes as a function on $\mathbb{F}_\ell$. At nonzero x this requires

$$x^{n-1} = 1 + a(x-1) \quad (x \in \mathbb{F}_\ell^\times).$$

Reducing X^{n-1} modulo $X^{\ell-1} - 1$, the left side is a single monomial of degree $0, \ldots, \ell - 2$, while the right side is a linear polynomial. Since evaluation on $\mathbb{F}_\ell^\times$ is injective for polynomials of degree at most $\ell - 2$, the exponent must reduce to 0 or 1 modulo $\ell - 1$. The case 0 is impossible because $n - 1$ is odd and $\ell - 1$ is even. Thus $\ell - 1 \mid n - 2$. Conversely that divisibility gives $x^{n-1} = x$ for every nonzero $x \in \mathbb{F}_\ell$ and $a \equiv 1 \pmod{\ell}$, so P_n vanishes. For $\ell = 3$ the same criterion holds directly: $2 \mid n - 2$ and $a \equiv 1 \pmod{3}$.

Suppose an odd ℓ divides d_r. Evaluating the polynomial at $1+\ell$ yields

$$P_n(1+\ell) \equiv \ell\,(n-2) \pmod{\ell^2},$$

so $\ell^2 \mid d_r$ forces $\ell \mid n-2$. Conversely, if both $\ell-1$ and ℓ divide $n-2$, Euler's theorem modulo ℓ^2 implies $x^{n-2} \equiv 1$ for every ℓ-adic unit, and also $a \equiv 1 \pmod{\ell^2}$. Consequently

$$P_n(x) = x^2(x^{n-2}-1) - (a-1)x(x-1) \equiv 0 \pmod{\ell^2}$$

for units, and separately for multiples of ℓ; hence $\ell^2 \mid d_r$. No odd ℓ^3 can divide d_r, because

$$P_n(2\ell) - 2P_n(\ell) \equiv -2a\ell^2 \not\equiv 0 \pmod{\ell^3},$$

where $a \equiv 1 \pmod{\ell}$. Finally $4 \mid P_n(x)$ for every integer x since n is even, $n \geq 4$, and $a \equiv -1 \pmod 4$; but $3!\left\{ {n \atop 3} \right\} = 3^n - 3\cdot 2^n + 3 \equiv 4 \pmod 8$, so $v_2(d_r) = 2$. Combining these local valuations proves (44). □

Remark 12.4 (Classical Stirling divisibility versus the CCR application)**.** The fixed-divisor object $\Delta_{n,3} = \gcd\{j!\left\{ {n \atop j} \right\} : 3 \leq j \leq n\}$ is *not* introduced here: Lundell studied prime-power divisibility of precisely $\Delta_{n,m}$ in 1978 [5]. Formula (44) is included with a self-contained elementary proof as a useful classical specialization, *not claimed as an independently original Stirling theorem.* The CCR-specific interpretation uses the previously established first-order source transform to identify the exact source image and its permitted-prime resonance consequences. In particular, $\ell \mid d_r$ occurs only when $\ell \leq r-2$; these potentially small primes need not satisfy the independent p-integrality hypotheses of Theorem 12.6, and no local splitting claim is inferred there merely from this divisibility statement.

Corollary 12.5 (All-odd-order fixed-source nonresonance)**.** *Choose the coherent integral stationary source family $\lambda_a = (-1)^{a-1}$ for every $a \geq 3$. At each fixed odd $r \geq 5$, and for* every *prime satisfying the order-r good-prime hypotheses of Theorem 12.6, the zero-residue fold has no roots modulo p^2 or over $\mathbb{Q}_p$ and is an irreducible totally ramified quadratic factor. There are no additional* source-dependent *exceptional primes for this family.*

Proof. Theorem 12.2 gives $A_r = 1$, which is a unit at every prime. The proof of Theorem 12.6 applies verbatim. □

The first new higher-order example beyond the fifth column is

$$\begin{aligned} A_5 &= 13 + 36\lambda_4 + 24\lambda_5, \\ A_7 &= 61 + 540\lambda_4 + 1560\lambda_5 + 1800\lambda_6 + 720\lambda_7. \end{aligned} \tag{45}$$

At order nine one obtains $A_9 = 253 + 5796\lambda_4 + 40824\lambda_5 + 126000\lambda_6 + 191520\lambda_7 + 141120\lambda_8 + 40320\lambda_9$.

Theorem 12.6 (Universal fixed-source finite prime-support theorem)**.** *Let $r = 2s+1 \geq 5$ be fixed and $\lambda_4, \ldots, \lambda_r \in \mathbb{Z}$. Choose a positive integer D_r clearing the rational coefficients of both S_r and E_r as polynomials in their independent variables, as guaranteed by stationary polynomiality. Put $c_s = -B_{2s}/(4s(2s)!) \neq 0$. At every prime p such that*

$$p > r, \qquad p \nmid D_r, \qquad v_p(c_s) = 0, \qquad p \nmid A_r(\boldsymbol{\lambda}),$$

the disk $m \equiv 0 \pmod p$ contributes exactly one double residual root modulo p, *but* no solutions modulo p^2, *no solutions modulo higher powers, and no $\mathbb{Q}_p$ roots. Its degree-two local factor is irreducible and totally ramified. Consequently the primes at which this CCR odd-column zero-disk fold can split form a* finite *set, contained in the union of the finite order-dependent bad-prime set and the prime divisors of the explicit nonzero integer $A_r(\boldsymbol{\lambda})$.*

Proof. At the stated primes, the established integral coefficientwise reduction gives $\Pi_{p,r} \equiv S_r \pmod{p}$. The classical Nörlund coefficient identity gives $S_r(m) = c_s m^2 + O(m^3)$, with c_s a p-unit, so the disk is a nondegenerate double residual root. For $m = pz$, $z \in \mathbb{Z}_p$, we have $S_r(pz) \equiv 0 \pmod{p^2}$, and polynomial integrality yields

$$\Pi_{p,r}(pz) \equiv pE_r(0, 0, \boldsymbol{\lambda}) = -\frac{pA_r(\boldsymbol{\lambda})}{r!} \not\equiv 0 \pmod{p^2}.$$

The stated conditions make $r!$ invertible and A_r a unit modulo p. Thus the whole disk vanishes at the next digit. The derivative-zero center in that disk exists uniquely by Hensel's lemma applied to the derivative; its critical value has p-valuation exactly one. Its quadratic Newton polygon therefore has slope $-1/2$, proving irreducibility and total ramification. The nonvanishing congruence (40) makes A_r a nonzero integer, hence it has finitely many prime divisors. □

Corollary 12.7 (Finite-prime simultaneous tuning versus fixed-source rigidity)**.** *Fix an odd order $r \geq 5$, fixed lower integer sources $\lambda_4, \ldots, \lambda_{r-1}$, and a finite set $\mathcal{P}$ of pairwise distinct order-r good primes. There exist* ordinary integer *top source values λ_r for which the zero-disk quadratic factor splits into two distinct $\mathbb{Q}_p$-linear factors* simultaneously for all $p \in \mathcal{P}$. *Precisely, $\prod_{p\in\mathcal{P}}(p-1)/2$ residue classes of λ_r modulo $\prod_{p\in\mathcal{P}} p^2$ realize splitting at the first resonance depth $v_p(\lambda_r - \Lambda_r) = 1$ at every such prime. Nevertheless for every fixed integral source vector the number of primes at which this quadratic factor can split is finite.*

Proof. At each good prime, the top-source translation $\Pi_{p,r}^{(\lambda_r)} = \Pi_{p,r}^{(0)} - p\lambda_r/r$ and the local normal form in Theorem 8.2 give a unique critical $\Lambda_r \in \mathbb{Z}_p$ and a unit K_r with $z^2 = K_r p(\lambda_r - \Lambda_r)$. Exactly one class modulo p satisfies $\lambda_r \equiv \Lambda_r \pmod{p}$. Of its p lifts modulo p^2, exactly $(p-1)/2$ satisfy $v_p(\lambda_r - \Lambda_r) = 1$ and $\left(\frac{K_r(\lambda_r-\Lambda_r)/p}{p}\right) = 1$. These lifts split as claimed. The Chinese remainder theorem combines independent prime-square conditions, giving the exact product count and integer representatives. For any one such integer representative, Theorem 12.6 limits the split primes to the fixed finite order-dependent bad-prime set and prime divisors of the nonzero integer A_r. □

Remark 12.8 (Exact finite example, order five)**.** With $\lambda_4 = 0$ and $\lambda_5 = 774982$, both zero-disk roots split at $p = 7, 11, 13$. The corresponding local field-root counts are three at each prime, whereas the unnormalized polynomial's full congruence-root counts for $n = 1, 2, 3, 4$ are, respectively,

$$p = 7 : \ (2, 8, 15, 15), \quad p = 11 : \ (2, 12, 23, 23), \quad p = 13 : \ (2, 14, 27, 27).$$

The difference between the field-root number and stable congruence-fiber cardinality is essential: a split local root with nonunit derivative represents p distinct classes modulo each sufficiently high power. The exact integer witness is a finite consistency check, not a substitute for the CRT proof.

Corollary 12.9 (Uniform source-height threshold)**.** *If $|\lambda_a| \leq B$ for all $4 \leq a \leq r$, then*

$$|A_r| \leq C_r(B) := 2^{r-1} - 3 + B \sum_{a=4}^{r} (a-1)! \left\{ {r-1 \atop a-1} \right\}.$$

For every prime meeting the order-r good-prime conditions and satisfying $p > C_r(B)$, the zero-residue disk disappears at precision p^2 uniformly for every integral source vector of that height. *For $r = 7$, $C_7(B) = 61 + 4620B$; the old $r = 5$ bound is $13 + 60B$.*

Remark 12.10 (Precise global scope and historical precedence)**.** Theorem 12.6 concerns the CCR-specific zero-residue quadratic factor. It does *not* assert an all-prime factorization of the remaining degree-$r-2$ factor for every $r \geq 7$. If $S_r(m)/m^2$ is squarefree modulo a particular good prime p, then all nonzero residual roots are simple and the complete root count modulo every p^n, $n \geq 2$, equals the number of nonzero roots of $S_r \pmod p$; in characteristic zero the quotient is explicitly squarefree for $r = 5, 7, 9, 11$ by finite symbolic calculation. The alternating Stirling sum and Bézout argument in Theorem 12.2 are classical combinatorial facts, not an assertion of new Stirling-number identities. Their interpretation as an exact CCR source-obstruction lattice and simultaneous p-adic splitting criterion is the qualified specialized contribution. The Nörlund identity, and the odd-index double-zero factor itself, precede the present work: see Bencherif–Garici [4]. The new proposed contribution is the *CCR source-deformation formula* (39) together with the parity obstruction and finite-prime-support application, subject to further full-text novelty review.

13 Chronological integer-source observations and their limits

The preceding stationary theory concerns a single vector of source parameters. It is important to ask exactly which *terminal raw coefficient* constraints survive when only the top order-r source varies in time. This section gives finite, closed criteria; its assertions are *not* identities of entire normalized polynomial columns, and there is no claim that the prescribed source amplitudes arise from a physical DPDA transition system.

Fix an odd integer $r \geq 5$. Retain the stationary lower integer sources $\lambda_3 = 1, \lambda_4, \ldots, \lambda_{r-1}$ and let the top source at temporal row $t \geq r$ be an independent integer $\eta_{r,t}$. Write

$$w_t = \binom{t-1}{r-1}, \quad B_N = \binom{N}{r} = \sum_{t=r}^{N} w_t, \quad Z_N = \sum_{t=r}^{N} w_t \eta_{r,t}, \quad \lambda_{\mathrm{eff}}(N) = \frac{Z_N}{B_N}. \tag{46}$$

The r-th connected source enters the raw CCR recurrence in degree r, so changes to its temporal sequence affect the degree-r coefficient linearly through the terminal-block weights w_t; no lower-degree coefficient is affected. Thus $\lambda_{\mathrm{eff}}(N)$ is the *unique stationary top-source shadow* of that single raw terminal coefficient. From Theorem 12.1 the first-order obstruction of this shadow is

$$A_r^{\mathrm{shadow}}(N) = L_r + (r-1)!\lambda_{\mathrm{eff}}(N), \quad L_r = 2^{r-1} - 3 + \sum_{a=4}^{r-1} (a-1)! \left\{ {r-1 \atop a-1} \right\} \lambda_a. \tag{47}$$

Because $L_r \equiv 1 \pmod 6$ and $(r-1)!$ is divisible by 6, the cancellation target $c_r = -L_r/(r-1)!$ is a noninteger rational.

Theorem 13.1 (Signed integer multisnapshot criterion)**.** *Let $r \leq N_1 < \cdots < N_m$, put $N_0 = r-1$, and fix $c \in \mathbb{Q}$. With unrestricted signed integer sources $\eta_{r,t}$, one history satisfies $\lambda_{\mathrm{eff}}(N_i) = c$ for all i if and only if*

$$c\left(\binom{N_i}{r} - \binom{N_{i-1}}{r} \right) \in g_i \mathbb{Z}, \qquad g_i = \gcd\{w_t : N_{i-1} < t \leq N_i\}, \quad 1 \leq i \leq m. \tag{48}$$

In particular no integer history can cancel A_r^{shadow} at two consecutive terminal rows, but there are single signed integer histories attaining the cancellation at infinitely many suitably separated rows.

Proof. The independent increment $Z_{N_i} - Z_{N_{i-1}}$ is an integer linear combination of the weights in that interval. Bézout's identity characterizes all such increments as $g_i\mathbb{Z}$; separate intervals

do not share source cells, giving necessity and sufficiency. For consecutive snapshots subtraction in (46) forces $\eta_{r,N+1} = c_r$, impossible since $c_r \notin \mathbb{Z}$. Conversely, the $(r-1)$-st finite difference gives $\gcd(w_{N+1}, \dots, w_{N+r}) = 1$. Writing $c_r = a/D$ in lowest terms, the sequence $N_j = j\, r!D$ has integral $c_r\binom{N_j}{r}$ and gaps at least r. Formula (48) then produces the claimed infinite history. □

Theorem 13.2 (Nonnegative integer sources and exact reachability)**.** *Keep the same lower stationary parameters, which may be signed, but impose $\eta_{r,t} \in \mathbb{Z}_{\geq 0}$. For the snapshot rows of Theorem 13.1, let $\mathcal{S}_i = \langle w_t : N_{i-1} < t \leq N_i \rangle_{\mathbb{Z}_{\geq 0}}$. One nonnegative integer history realizes the rational target c at all snapshots if and only if*

$$c\left(\binom{N_i}{r} - \binom{N_{i-1}}{r}\right) \in \mathcal{S}_i \qquad (1 \leq i \leq m). \tag{49}$$

For $c < 0$ no snapshot is attainable; for $c = 0$ all are attainable using zero sources; for every $c > 0$ one history attains infinitely many sparse snapshots, without a uniform amplitude or gap bound.

Proof. Equation (49) says exactly that each independent nonnegative weighted increment is representable by source pulses in its time interval. Negative increments are impossible and zero increments are attained by the zero history. For $c = a/D > 0$, choose an initial large multiple N_1 of $r!D$ and place the integral amount cB_{N_1} at weight $w_r = 1$. Given an attained N_j, the first r weights past N_j have gcd one by finite differences. Their nonnegative numerical semigroup has a finite complement in $\mathbb{Z}_{\geq 0}$ [21]. Choose the next sufficiently large multiple N_{j+1} of $r!D$ so that the integral required increment $c(B_{N_{j+1}} - B_{N_j})$ lies beyond the conductor. Represent it using those first r weights and set all other intervening sources to zero. This proves induction. □

Theorem 13.3 (Sharp mixed-sign lower-band threshold)**.** *Let all $\eta_{r,t} \in \mathbb{Z}_{\geq 0}$ remain unrestricted in amplitude. Define positive Stirling weights and the signed lower-source budgets*

$$K_{r,a} = (a-1)!\left\{ {r-1 \atop a-1} \right\} > 0, \quad P_r = \sum_{a=4}^{r-1} K_{r,a} \max(\lambda_a, 0), \qquad D_r = \sum_{a=4}^{r-1} K_{r,a} \max(-\lambda_a, 0).$$

Then $A_r^{\mathrm{shadow}}(N) = 0$ is attainable for some terminal row N if and only if

$$\boxed{D_r \geq 2^{r-1} + 2 + P_r.} \tag{50}$$

For each odd $r \geq 5$, the separate *choice $\lambda_4 = -1$ and $\lambda_5 = \cdots = \lambda_{r-1} = 0$ satisfies this threshold. In the typed higher-band Hessenberg realization, writing $a_{r,t} = (-1)^r p \eta_{r,t}$ gives*

$$H_{i+r-1,i} = -p\, i(i+1)\cdots(i+r-2)\eta_{r,i+r-1};$$

thus a negative lower stationary λ_a is equivalent to a positive *corresponding lower Hessenberg entry. This equivalence is formal and does not assert DPDA realizability.*

Proof. By (47), $L_r = 2^{r-1} - 3 + P_r - D_r \equiv 1 \pmod 6$. Nonnegative top sources imply $\lambda_{\mathrm{eff}}(N) \geq 0$, so cancellation requires $L_r < 0$, equivalently (50); equality $L_r = 0$ is excluded by its congruence. Conversely if $L_r < 0$, set $c = -L_r/(r-1)! = a/D > 0$ in lowest terms and choose $N = r!D$. Since $D \mid \binom{N}{r}$, the single nonnegative source pulse $\eta_{r,r} = c\binom{N}{r}$ attains the required shadow. Finally $K_{r,4} = 3^{r-1} - 3 \cdot 2^{r-1} + 3$, so the stated lower choice gives $L_r = 4 \cdot 2^{r-1} - 3^{r-1} - 6 < 0$ for odd $r \geq 5$. The Hessenberg identity is the explicit signed source realization of the established typed CCR recurrence. □

Remark 13.4 (Interpretation and a checkable example)**.** At $r = 5$, $\lambda_4 = -1$ gives $L_5 = -23$ and $c_5 = 23/24$. The source $\eta_{5,5} = 759$ yields the target at $N = 12$, as $759/\binom{12}{5} = 23/24$. It cannot simultaneously be attained at $N = 16$ with nonnegative pulses confined to the interval $(12, 16]$, although that interval has weights of gcd one; 3427 is outside the semigroup generated by $495, 715, 1001, 1365$. At $(12, 18]$ the pulses $\eta_{5,13} = 3$, $\eta_{5,14} = 3$, $\eta_{5,15} = 2$, and $\eta_{5,17} = 1$ give the required increment 7452. This separates signed Bézout reachability from nonnegative semigroup reachability. The comparison remains at *one terminal raw degree-r coefficient*, not at the level of full p-adic normalized columns.

14 Explicit Galois laws at the seventh and ninth columns

The odd-column result above removes the double zero residue disk, but it leaves the nonzero residual roots to be determined. Orders seven and nine provide complete global root distributions at almost every prime, as an arithmetic consequence of the *previously known* Stirling polynomials. Bencherif–Garici [4] explicitly list the underlying quartic and sextic (with the opposite overall sign). We neither claim their polynomial formulas nor general Chebotarev theory as new. The present statement connects their Frobenius cycle types to *fixed-source CCR* prime-power root counts by Theorem 12.6.

Write

$$q_7(m) = 9m^4 - 54m^3 + 51m^2 + 58m + 16, \tag{51}$$

$$q_9(m) = 15m^6 - 165m^5 + 465m^4 + 17m^3 - 648m^2 - 548m - 144. \tag{52}$$

The Nörlund identities already used in Section 8 give

$$S_7(m) = \frac{m^2(m-1)q_7(m)}{5806080}, \qquad S_9(m) = \frac{m^2(m-1)q_9(m)}{2786918400}. \tag{53}$$

The factors $m - 1$ are simple, and the two quotients are squarefree over $\mathbb{Q}$: directly,

$$\operatorname{disc}(q_7) = -2^4 3^5 \cdot 83 \cdot 97 \cdot 271,$$
$$\operatorname{disc}(q_9) = -2^{16} 3^3 5^2 \cdot 7 \cdot 263 \cdot 67891 \cdot 63049079.$$

Furthermore $q_7(0) = 16$, $q_7(1) = 80$, $q_9(0) = -144$ and $q_9(1) = -1008$, so these extra roots do not collide with 0 or 1 for sufficiently large primes away from the displayed discriminants.

Theorem 14.1 (Exact small-order Galois groups)**.** *For the explicit polynomials* (51)–(52)*,*

$$\operatorname{Gal}(q_7/\mathbb{Q}) \cong S_4, \qquad \operatorname{Gal}(q_9/\mathbb{Q}) \cong S_6.$$

Proof. The squarefree modular factor-degree certificates for q_7 are (4) at $p = 13$ and $(1, 3)$ at $p = 17$. Thus the Galois group is transitive and contains both a four-cycle and a three-cycle. The only transitive subgroup of S_4 with both types is S_4.

For q_9, the factor-degree certificates are (6) at $p = 29$, $(1, 5)$ at $p = 17$, and $(1, 1, 1, 1, 2)$ at $p = 607$. None of these primes divides $\operatorname{disc}(q_9)$. Consequently the Galois group $G \leq S_6$ is transitive and contains a five-cycle and a transposition. It is primitive: a nontrivial system of blocks in a transitive degree-six action has block size two or three, and a five-cycle cannot act nontrivially on either the blocks or any such block. Finally, the graph on six letters whose edges are the G-conjugates of the supplied transposition has a G-invariant component partition. Primitivity forces that graph to be connected; its edge transpositions generate S_6. Hence $G = S_6$. The stated factor degrees are determined by factoring the explicit integer polynomials (51)–(52) in the indicated finite fields. The discriminants follow by exact resultant evaluation, $\operatorname{disc}(q) = (-1)^{d(d-1)/2} \operatorname{Res}(q, q')/\operatorname{lc}(q)$ for $d = \deg q$. Thus the cycle-type witnesses and discriminant assertions are finite, reproducible algebraic certificates rather than additional hypotheses. □

For each odd r, let $A_r(\boldsymbol{\lambda})$ be the nonzero integer in (39). Define the two explicit finite bad-prime sets

$$\mathcal{B}_7(\boldsymbol{\lambda}) = \{p \text{ prime} : p \le 14 \text{ or } p \mid A_7(\boldsymbol{\lambda})\, 83 \cdot 97 \cdot 271\},$$
$$\mathcal{B}_9(\boldsymbol{\lambda}) = \{p \text{ prime} : p \le 18 \text{ or } p \mid A_9(\boldsymbol{\lambda})\, 263 \cdot 67891 \cdot 63049079\}.$$

Here the small-prime cutoffs also ensure that every denominator in the order-r normalized transport is p-invertible. Indeed the triangular operator T_j in Section 12 has diagonal entries $j+d$ for $0 \le d \le j$, all at most $2r$, and its right-hand sides have integral coefficients once the earlier columns are integral. Thus, by induction, the coefficients of S_r and its p-deformation are p-integral for all $p > 2r$. No claim is made at the explicitly excluded primes.

Theorem 14.2 (Fixed-source seventh/ninth prime-axis laws)**.** *Fix a stationary integer CCR source vector $\lambda_3 = 1, \lambda_4, \ldots, \lambda_r \in \mathbb{Z}$ and $r \in \{7, 9\}$. For any rational prime $p \notin \mathcal{B}_r$, and every $n \ge 2$, the number of solutions of the normalized r-th CCR column modulo p^n equals*

$$1 + \#\{u \in \mathbb{F}_p : q_r(u) = 0\},$$

and is also the number of distinct roots in $\mathbb{Q}_p$. Their natural densities, among all rational primes, are

root count $(r=7)$	1	2	3	5
density	9/24	8/24	6/24	1/24

root count $(r=9)$	1	2	3	4	5	7
density	265/720	264/720	135/720	40/720	15/720	1/720.

In both cases the prime-axis mean number of distinct roots converges to 2, independently of the fixed integral source vector.

Proof. For $p > 2r$, the original normalized transport reduces coefficientwise to S_r modulo p. By (53) and the exclusion of $\mathcal{B}_r$, every nonzero residual root comes either from $m = 1$ or a simple root of q_r and lifts uniquely modulo all p^n by Hensel's lemma. The double root at zero contributes no solution modulo p^2, by Theorem 12.6, because $p \nmid A_r$. This proves the exact count for every such prime, not merely a statistical limit.

Apply Chebotarev to the groups in Theorem 14.1. The number of roots of q_r modulo an unramified prime is the number of fixed points of the corresponding permutation. For S_4, the permutation counts with $0, 1, 2, 4$ fixed points are $9, 8, 6, 1$ out of $4! = 24$. For S_6, the counts with $0, 1, 2, 3, 4, 6$ fixed points are $265, 264, 135, 40, 15, 1$ out of $6! = 720$ (the classical rencontres numbers $\binom{6}{k} D_{6-k}$). The unavoidable root $m = 1$ adds one to each. The finite excluded sets do not change the natural densities. In both symmetric groups the mean number of fixed points is one, so the total mean is two. □

Corollary 14.3 (Conditional all-odd orbit-count mean)**.** *Fix any odd $r \ge 5$. Over $\mathbb{Q}$, factor $S_r(m) = m^2(m-1)q_r(m)$ with $q_r \in \mathbb{Q}[m]$. Assume q_r is squarefree and $q_r(1) \ne 0$, and let t_r be its number of distinct irreducible factors over $\mathbb{Q}$. For any fixed integral stationary source vector, the mean number of $\mathbb{Q}_p$ roots of the normalized r-th column, averaged over rational primes $p \le X$, tends to $1 + t_r$ as $X \to \infty$. The same mean holds for the number of roots modulo p^n for every fixed $n \ge 2$ outside finitely many source-dependent primes. In particular, if q_r is irreducible, the mean is exactly 2.*

Proof. The classical odd Nörlund factorization and its exact double zero at the origin are already cited in Section 8; the factor $m - 1$ follows from vanishing odd Bernoulli numbers at $m = 1$. For a fixed r, away from finitely many coefficient-denominator, discriminant, resultant and A_r prime divisors, all nonzero residual roots are simple and lift, whereas the

zero disk disappears modulo p^2 by Theorem 12.6. The remaining roots are the single root at $m = 1$ and the fixed points of Frobenius acting on the roots of q_r. By the Chebotarev density theorem the limiting average number of such fixed points is the group-average number of fixed points. Burnside's lemma identifies it with the number of Galois orbits, namely t_r, the number of irreducible factors of the squarefree rational polynomial q_r. The finite excluded prime set has zero density. □

Remark 14.4 (Exactly what is new here)**.** The explicit q_7, q_9 appeared in the 2017 Stirling-polynomial literature [4]. Their algebraic Galois computations and the Frobenius fixed-point technique are standard mathematical approaches. The *CCR-specific arithmetic result* of this section is the combination with the previously established integral source obstruction A_r, producing source-independent global p-adic root-count laws at all but finitely many primes. This is an arithmetic specialization, not a theorem about all odd r or actual DPDA source realizability; the underlying unperturbed polynomial identities are properly attributed to the Stirling literature.

15 Relation to earlier work and theorem provenance

A source-level audit is important here because the integer polynomial families S_r arise from classical Nörlund theory. In particular, Adelberg's studies [6, 7] cover Bernoulli–Stirling polynomials and congruences, and the latter determines sharp coefficient denominators for classical Nörlund polynomials. Consequently neither the formal $p = 0$ generating function, the unperturbed denominator behavior, nor the mere existence of a Stirling polynomial can be attributed to CCR-k. Bencherif and Garici [4] establish the odd-index vanishing pattern and explicitly print the polynomials whose signs give q_7 and q_9. These comparisons rule out presenting those ingredients as original results of this manuscript. More precisely, the fixed divisor $\Delta_{n,3} = \gcd_{3\le j\le n} j!\left\{ {n \atop j} \right\}$ was already studied by Lundell [5]; its valuation theory is not attributed to CCR-k. The arithmetic claim made here is the *source-image identification* in Theorem 12.2, $A_r(\boldsymbol{\lambda}) \in 1 + d_r\mathbb{Z}$ with exact image, and its fixed-integral-source consequence in Theorem 12.6. These claims rely on the *nonclassical input of the specified CCR transport*, not merely on a formula for $\Delta_{n,3}$. The prime-factor expression for d_r is given with an independent proof for completeness, not advertised as an independently new fixed-divisor theorem.

The double-discriminant factorization into a squared Maxwell component and a cubed caustic component is a general polynomial phenomenon [16]; its presence alone is not a newly discovered CCR mechanism. Likewise generic symmetric Galois groups in polynomial families with two free coefficients have classical theorems of Cohen and the subsequent correction [17, 18]; Morse critical-point monodromy is classical [20]. Recent work on critical values includes [19], which treats equicritical quartics rather than the particular CCR fifth-column integer coefficients considered here. The exact CCR eliminants, source-depth certificates and arithmetic specializations remain substantive project-specific calculations, but the present conclusions do not include any new general Maxwell/caustic or two-coefficient Galois theorem. In particular, the generic two-parameter S_5 conclusion is *not* a claimed new theorem about Galois groups: it is an instance of characteristic-zero existing theory. The distinct assertion available here is the integer/73-adic decision for this *particular normalized CCR quintic*, including its critical-value-depth and local splitting certificates. The same distinction applies to the 419 quartic: the original quartic formula precedes this refinement; the finite integer discriminant certificate and its local ramification classification are the statements proved here.

The distinction among the mathematical ingredients is summarized below.

Mathematical ingredient	Provenance and present status
Typed CCR transport and quartic $G_{p,\lambda}$	Previous CCR work; standing input, not a theorem newly established here.
Nörlund series and odd m^2 zero; polynomials q_7, q_9	Classical; see [6, 7, 4].
Hensel, Newton polygons, local square coordinates, Haar counting, Chebotarev, Burnside	Established tools; specializations and computations only.
Exact CCR source image $A_r(\boldsymbol{\lambda}) \in 1 + d_r\mathbb{Z}$; fixed-integer zero-fold finite prime support	CCR-specific source-image and resonance consequences are established using the classical Stirling fixed divisor; the fixed divisor itself is not new.
Signed/positive terminal shadow reachability; exact band budget	Classical Bézout, finite-difference and numerical-semigroup tools applied to CCR weighted sources; scope is strictly temporal raw coefficient, not physical DPDA.
Prime 419 finite discriminant certificate; two folds at 73 and joint source behavior	Explicit integer certificates and local CCR-family arithmetic formulas established for these polynomials; general critical-value methods are classical.
S_4/S_6 Galois and prime densities for $r = 7, 9$	Factor/cycle certificates for previously published underlying polynomials; CCR-specific consequences concern fixed-source arithmetic.

For the S_4 and S_6 claims, independent integer-polynomial verification also confirms both discriminants, and factor-degree witnesses at primes $13, 17$ (degree four) and $17, 29, 607$ (degree six), respectively. The primitive integral cubic used in the 419-certificate has no root modulo 23, and modulo 419 reduces to a linear polynomial with unit slope. These finite verifications validate the displayed certificates; they do not amount to a full independent proof check of all the p-adic classification theorems or establish bibliographic uniqueness.

Scope of the arithmetic contribution. The mathematical claims of this refinement concern the exact CCR source perturbation, its integer-image and resonance obstructions, and the stated arithmetic classifications of its normalized polynomial families. The general Nörlund, finite-difference, discriminant, Galois and probabilistic tools are attributed to their antecedents. The CCR predecessor references are identified below. None of the results asserts that arbitrary typed source coefficients are physically realized by a DPDA.

16 Exact-arithmetic checks for the critical-prime models

The following finite-depth congruence certificates are obtained by reducing the displayed integer coefficient formulas modulo prime powers, lifting derivative-critical residues by Hensel's lemma, and counting the resulting congruence classes. They supplement, but do not replace, the exact all-depth proofs in Lemma 3.1 and Theorem 9.2.

The 419**-adic coordinate and its domain.** A digitwise Hensel lift of the derivative-zero center followed by evaluation of the critical value gives the first six base-419 digits of the critical parameter as

$$\Lambda \equiv 132 + 91 \cdot 419 + 73 \cdot 419^2 + 157 \cdot 419^3 + 402 \cdot 419^4 + 44 \cdot 419^5 \pmod{419^6}.$$

For $\lambda = 0, 132, 551, 38261$, direct congruence lifting from the integer quartic, without using the closed solution-count formula, gives the following root counts:

λ	N_1	N_2	N_3
0	3	2	2
132	3	421	2
551	3	421	840
38261	3	421	421

The isometry claim concerns integral residue classes. As clarified in Lemma 3.1, half-integral-valuation algebraic roots are instead classified by the Newton polygon of the translated integral polynomial.

The 73-adic covariance. For the fifth-column quintic, reducing the original integer polynomial modulo 73, 73^2, and 73^3 and averaging over all $73^2 = 5329$ source residues $T \pmod{73^2}$ for fixed L gives

L	$d = v_{73}(\Lambda_{62} - \Lambda_0)$	$\mathbb{E}_T[M_{0,3}M_{62,3}]$	$\mathrm{Var}_T(N_3)$
0	0	0	286
9	1	72	430
812	2	145	576

Each individual fold has mean 1. These counts independently validate all three regimes of Theorem 9.2 at precision $n = 3$. For clarity, the *ordered* critical gap throughout this paper is $\delta = \Lambda_{62} - \Lambda_0$, and its first residue is $\delta \equiv -2 - 16L \pmod{73}$. The opposite sign refers to the reverse ordered gap and must not be substituted into the product congruence without also reversing its sign. All formulas and independent counts in this article use this ordered gap convention; reversing the order also reverses the sign in the corresponding product congruence.

Novelty boundary. Analytic Morse coordinates, Hensel lifting, and such finite modular counts are established tools, not claimed as new [12]. The finite congruence certificates concern precisely the displayed prime powers and parameter classes; the general statements rest on the separate all-depth arguments.

17 Haar-product independence across different odd CCR columns

The preceding fifth-column two-fold theorem couples two critical disks *inside one column* at the exceptional prime 73. We now consider one distinguished zero-residue fold in *each of several different odd columns*, all at the same prime. This is a separate assertion whose proof uses the original CCR-k triangular source structure [3], not the special 73-adic two-curve isometry.

Normalized p-integrality at large primes. The good-prime hypothesis below can be checked directly from the original stationary transport, rather than assumed implicitly. For $1 \le j \le R$ and $0 \le k \le j$, its triangular operator satisfies

$$T_j(m^k) = (j+k)m^k + \sum_{h=0}^{k-2} \binom{k}{h} m^{h+1},$$

where an empty sum means zero. On polynomials of degree at most j, T_j has integral triangular matrix and diagonal $j, j+1, \ldots, 2j$. If $p > 2R$, all these diagonal entries are

p-adic units, so T_j^{-1} preserves the module of p-integral polynomials of degree at most j. The original normalized CCR recurrence has p-integral polynomial right-hand side whenever the preceding normalized columns and the stationary sources are p-integral. Induction from $\Pi_{p,0} = 1$ therefore shows that every $\Pi_{p,j}$, $j \le R$, is coefficientwise p-integral for every source vector in $\mathbb{Z}_p$; specialization at $p = 0$ gives $\Pi_{p,j} \equiv S_j \pmod p$. The additional unit-curvature condition stated below ensures that the residual zero-disk double root is nondegenerate. No assertion is made for the finitely many smaller or curvature-bad primes.

Theorem 17.1 (Odd-column Haar product law)**.** *Fix a finite set $I \subseteq \{5, 7, 9, \dots, R\}$ of odd indices and put $d = |I|$. Let $p > 2R$ be an odd prime such that, for every $r = 2s+1 \in I$, the curvature*

$$c_s = -\frac{B_{2s}}{4s(2s)!} \in \mathbb{Z}_p^\times.$$

Work with the stationary normalized CCR polynomial extension, $\lambda_3 = 1$, and regard all source parameters with indices outside I as arbitrary fixed elements of $\mathbb{Z}_p$. Choose $\{\lambda_r : r \in I\}$ independently with normalized Haar measure on $\mathbb{Z}_p$. For $n \ge 1$, denote by $M_{r,n}$ the number of zeros of $\Pi_{p,r}(m)$ modulo p^n with $m \equiv 0 \pmod p$.

Then the variables $M_{r,n}$, $r \in I$, are independent and have the same single-fold law. More precisely, define

$$h = \left\lfloor \frac{n-1}{2} \right\rfloor, \quad q = p^{-(n-1)}, \quad a = p^{\lfloor n/2 \rfloor}, \quad w_j = \frac{p-1}{2p^{2j}}$$

and the probability polynomial

$$P_{p,n}(X) = \left(1 - q - \sum_{j=1}^{h} w_j\right) + qX^a + \sum_{j=1}^{h} w_j X^{2p^j}. \tag{54}$$

For $S_n = \sum_{r \in I} M_{r,n}$,

$$\mathbb{E}[X^{S_n}] = P_{p,n}(X)^d, \qquad \mathbb{E}[S_n] = d, \qquad \mathrm{Var}(S_n) = d(p-1)(n-1). \tag{55}$$

The number B of indices in I whose zero-residue quadratic factor has two distinct *$\mathbb{Q}_p$-roots satisfies*

$$B \sim \mathrm{Binomial}\left(d, \frac{1}{2(p+1)}\right). \tag{56}$$

The possible repeated-root critical parameters form a Haar-null set.

Proof. For each $r \in I$, the established normalized shadow has $S_r(m) = c_s m^2 + O(m^3)$ and a unit quadratic coefficient modulo p. The p-integral transport admits a unique derivative-critical center $c_r(\lambda_{<r}) \in p\mathbb{Z}_p$. This center depends only on the lower-order sources, because the already established exact stationary translation law for the top source is

$$\Pi_{p,r}^{(\lambda_r)}(m) = \Pi_{p,r}^{(0)}(m) - \frac{p}{r}\lambda_r$$

for odd r, with all lower sources fixed. Define $\Lambda_r(\lambda_{<r}) \in \mathbb{Z}_p$ by $\Pi_{p,r}^{(\Lambda_r)}(c_r) = 0$. Since the critical value of $\Pi^{(0)}$ is divisible by p and r is invertible, such a parameter exists uniquely. The elementary p-adic quadratic coordinate in the disk $m \in p\mathbb{Z}_p$ produces an isometric coordinate $z_r \in p\mathbb{Z}_p$ satisfying

$$z_r^2 = K_r p U_r, \qquad U_r = \lambda_r - \Lambda_r(\lambda_{<r}), \qquad K_r = \frac{1}{rc_s} \in \mathbb{Z}_p^\times. \tag{57}$$

Here the analytic square-root normalization is applied only on the *integral residue disk*; field-factor ramification outside this disk, if any, follows separately from its Newton polygon.

The source substitution

$$(\lambda_4,\dots,\lambda_R)\longmapsto(\widetilde{\lambda}_4,\dots,\widetilde{\lambda}_R),\quad \widetilde{\lambda}_r=\begin{cases}\lambda_r-\Lambda_r(\lambda_{<r}), & r\in I,\\ \lambda_r, & r\notin I\end{cases}$$

is triangular with unit diagonal, and is invertible recursively: recover $\lambda_4,\lambda_5,\dots$ in increasing order. All critical functions are p-integral and compatible with reduction modulo every p^a by Hensel's implicit-function theorem. Hence this map is a bijection on each finite residue vector space and preserves product Haar measure. With parameters outside I fixed, the same recursive argument gives independent Haar U_r for $r\in I$.

The zero-disk count $M_{r,n}$ is now a function only of U_r via (57). Multiplication by each fixed unit K_r does not change a Haar-uniform distribution. Elementary quadratic valuation counting gives probability q for the fully resonant count a, probability w_j for a split count $2p^j$ at each $1\le j\le h$, and the remaining mass at zero. This is (54). In particular, $\mathbb{E}M_{r,n}=1$ and $\mathrm{Var}(M_{r,n})=(p-1)(n-1)$, the universal one-fold moment formulas already established in Section 8. Independence proves (55).

Finally, a quadratic branch splits with two distinct $\mathbb{Q}_p$ roots exactly when $v_p(U_r)=2j-1$ for some $j\ge1$, and the leading unit of K_rU_r is a square. Its probability is

$$\sum_{j\ge1}\frac{p-1}{2p^{2j}}=\frac{1}{2(p+1)}.$$

The branch events are independent, proving (56). □

Remark 17.2 (What is and is not independent)**.** Even when the lower sources are correlated through the higher transport, the upper-source coordinates give exact independence of these *distinguished local folds*. Nothing here asserts independence of the *full* root counts from all other residue disks, or of arbitrary nonstationary/physically realized source data. Triangular Haar changes and one-dimensional quadratic congruence counting are classical methods; the precise CCR application follows from the pre-existing top-source translation theorem.

Finite-precision two-column illustration. Direct polynomial reduction using the full normalized CCR transport (rather than the local normal form (57)) fixes $\lambda_4=\lambda_6=0$ and enumerates every $(\lambda_5,\lambda_7)\in\mathbb{F}_p^2$ for $p=17,19,23$. For every $m\in p\mathbb{Z}/p^2\mathbb{Z}$ the resulting joint distribution shows that $(M_{5,2},M_{7,2})$ takes values $(0,0),(p,0),(0,p),(p,p)$ with multiplicities $(p-1)^2,p-1,p-1,1$, respectively. These three complete finite-precision distributions are consistent with the independent all-depth triangular/Hensel proof.

18 Adelic resonance dichotomy: fixed integer sources versus profinite Haar sources

The all-odd fixed-integer source obstruction and the multicolumn Haar-product theorem have complementary quantifiers. The former fixes an integer vector and varies the prime; the latter varies source coordinates in a single p-adic space. They admit an exact common adelic formulation, provided these two ambient spaces are not conflated.

Fix a nonempty finite set of odd column orders $I\subseteq\{5,7,\dots,R\}$ and write $d=|I|$. Let $\mathcal{G}_R$ be the cofinite set of primes $p>2R$ at which all selected normalized columns are

coefficientwise p-integral and their quadratic curvatures $c_{(r-1)/2}$ are p-units. This exclusion set depends on R, not on the source parameters. Define the profinite source space

$$\widehat{\mathbb{Z}}^{R-3} = \prod_p \mathbb{Z}_p^{R-3},$$

with independent normalized Haar factors and $\lambda_3 = 1$. For $p \in \mathcal{G}_R$ and $r \in I$, let $E_{p,r}$ be the event that the distinguished $m \equiv 0 \pmod p$ fold of the r-th CCR column has two distinct $\mathbb{Q}_p$ roots. No condition is placed on other root residue disks.

Theorem 18.1 (Adelic CCR prime-by-column product law)**.** *On the above profinite Haar source space, the events $\{E_{p,r} : p \in \mathcal{G}_R,\ r \in I\}$ are mutually independent, with*

$$\mathbb{P}(E_{p,r}) = q_p := \frac{1}{2(p+1)}. \tag{58}$$

Writing $N_I(X) = \sum_{\substack{p \le X \\ p \in \mathcal{G}_R}} \sum_{r \in I} \mathbf{1}_{E_{p,r}}$, we have

$$\mu_I(X) := \mathbb{E}N_I(X) = \frac{d}{2} \log\log X + O_R(1), \tag{59}$$

$$\mathrm{Var}(N_I(X)) = \frac{d}{2} \log\log X + O_R(1). \tag{60}$$

Moreover, almost surely,

$$\frac{N_I(X)}{\frac{d}{2}\log\log X} \longrightarrow 1, \tag{61}$$

and the centered count satisfies the standard central limit theorem

$$\frac{N_I(X) - \mu_I(X)}{\sqrt{\mathrm{Var}(N_I(X))}} \implies \mathcal{N}(0,1).$$

For some positive constant $C_{I,R}$ (depending on the finite excluded prime set),

$$\mathbb{P}(N_I(X) = 0) = \prod_{\substack{p \le X \\ p \in \mathcal{G}_R}} (1 - q_p)^d \sim \frac{C_{I,R}}{(\log X)^{d/2}}. \tag{62}$$

In particular, almost every profinite source vector has infinitely many split zero-disk CCR folds in each *chosen odd column.*

Proof. At each fixed good prime, the multicolumn triangular coordinates of the construction in Section 17 express the critical source difference as $U_r = \lambda_r - \Lambda_r(\lambda_{<r})$. As the chosen source coordinates vary independently in $\mathbb{Z}_p$, the U_r are independent Haar-uniform. The local quadratic equation $z_r^2 = K_r p U_r$ has two distinct $\mathbb{Q}_p$ roots precisely when $v_p(U_r)$ is positive odd and the corresponding leading unit satisfies the quadratic-character condition. Since every K_r is a unit, the exact probability is

$$\sum_{j \ge 1} \frac{p-1}{2p^{2j}} = \frac{1}{2(p+1)}.$$

The profinite Haar space is the product of the local source spaces *over primes*, so the already-proved within-prime independence extends to the entire prime-by-column family.

The reciprocal-prime Mertens estimate [14] and $q_p = \frac{1}{2p} + O(p^{-2})$ imply $\sum_{p\in\mathcal{G}_R,\ p\le X} q_p = \frac{1}{2}\log\log X + O_R(1)$. The sum of q_p^2 converges, proving the first two asymptotics. For completeness, enumerate the independent indicators as Y_j, put $\mu_n = \sum_{j\le n}\mathbb{E}Y_j \to \infty$, and observe

$$\sum_j \frac{\mathrm{Var}(Y_j)}{\mu_j^2} \le \sum_j \frac{\mathbb{E}Y_j}{\mu_j^2} < \infty,$$

where the initial finite terms are ignored and the final convergence follows by comparing increments of μ_j with $u^{-2}du$. Kolmogorov's convergence theorem and Kronecker's lemma then give $\sum_{j\le n}(Y_j - \mathbb{E}Y_j)/\mu_n \to 0$ almost surely. Because the centered indicators satisfy $|Y_j - \mathbb{E}Y_j| \le 1$ while their cumulative variance tends to infinity, Lindeberg's condition eventually has no nonzero summands, so the stated central limit theorem follows.

The constant in the no-splitting asymptotic has an explicit convergent Euler-product expression. Put $\mathcal{B}_R = \{p : \ p \notin \mathcal{G}_R\}$ and $a_p = (1-q_p)/(1-1/p)^{1/2}$ for $p \in \mathcal{G}_R$. Since $\log a_p = O(p^{-2})$, the product $\prod_{p\in\mathcal{G}_R} a_p$ converges to a strictly positive value. The prime-product form of Mertens' theorem gives

$$C_{I,R} = e^{-d\gamma/2}\prod_{p\in\mathcal{B}_R}(1-p^{-1})^{-d/2}\prod_{p\in\mathcal{G}_R}\left(\frac{1-1/(2(p+1))}{\sqrt{1-p^{-1}}}\right)^d > 0,$$

where γ is Euler's constant. This supplies (62) without leaving the positive constant implicit. Applying the almost-sure conclusion to each of the finitely many individual columns proves the final assertion. □

Theorem 18.2 (Integral-source finiteness and the height-limit dichotomy)**.** *Fix all unselected source coordinates as ordinary integers, and let the selected coordinates λ_r range over ordinary integers. For* every fixed *integer source vector, only finitely many pairs (p,r) with $p \in \mathcal{G}_R$, $r \in I$ can satisfy $E_{p,r}$. In sharp contrast, let selected integer coordinates be chosen independently and uniformly in $[-H,H]\cap\mathbb{Z}$ and write $N_I(\infty)$ for their total number of split good-prime folds. Then, for every fixed nonnegative integer K,*

$$\lim_{H\to\infty}\mathbb{P}_H(N_I(\infty)\le K) = 0. \tag{63}$$

Nevertheless $N_I(\infty)$ is finite for every *integer source vector, including all those sampled at height H.*

Proof. For any odd order $r \in I$, the CCR first-order source obstruction $A_r(\boldsymbol{\lambda})$ belongs to $1 + d_r\mathbb{Z}$, so it cannot vanish for integral sources (in particular $A_r \equiv 1 \pmod 6$). The fixed-source theorem proves that a good-prime splitting implies $p \mid A_r(\boldsymbol{\lambda})$, giving finitely many such primes for each fixed integral source vector.

For a *fixed finite* collection of good primes and column orders, Chinese-remainder equidistribution of integer source vectors in large boxes gives the same joint limiting distribution as the product Haar source measure. Indeed, each local split event is a union of p-adic residue cylinders, and its only possible boundary lies on the critical-value equality locus $U_r = 0$, of Haar measure zero. Thus its frequency among integer sources can be approximated from above and below by finite residue conditions. For fixed X, the law of $N_I(X)$ under the integer-box model therefore converges to the finite-prime law in Theorem 18.1. Since always $N_I(\infty) \ge N_I(X)$,

$$\limsup_{H\to\infty}\mathbb{P}_H(N_I(\infty)\le K) \le \mathbb{P}_{\mathrm{Haar}}(N_I(X)\le K).$$

In fact this limit can be obtained without invoking an additional almost-sure theorem: whenever $\mu_I(X) > K$, Chebyshev's inequality and the exact mean/variance estimates yield

$$\mathbb{P}_{\mathrm{Haar}}(N_I(X)\le K) \le \frac{\mathrm{Var}(N_I(X))}{(\mu_I(X)-K)^2} = O_{I,R,K}\left(\frac{1}{\log\log X}\right).$$

Because X is arbitrary after taking the height limit, the preceding lim sup is zero, proving (63). This does not claim any uniform bound in H and X simultaneously. □

Remark 18.3 (Quantifiers and historical scope)**.** The integer lattice is countable and Haar-null inside the profinite source space. It is therefore essential *not* to conclude from the adelic almost-sure statement that any fixed ordinary integer source has infinitely many resonance primes. The derived zero–one, strong-law, central-limit, and Mertens-product arguments use classical probability and analytic number theory; their proposed CCR contribution is solely the specific source-dependent resonance events to which the product laws apply. Neither a general full-root-count independence claim nor a theorem of physical DPDA source realization is intended.

Exact small-prime CRT certificate. For the fifth column with $\lambda_4 = 0$, reduce the integer numerator modulo p^3 and classify all $\lambda_5 \pmod{p^2}$ for $p = 7, 11, 13$ by the number of roots with $m \equiv 0 \pmod{p}$. The number of first-depth splitting source classes is $(p-1)/2 = 3, 5, 6$, respectively, with the unique critical class giving p roots and the nonsquare classes giving none. CRT therefore gives $3 \cdot 5 \cdot 6 = 90$ simultaneously splitting classes modulo $(7 \cdot 11 \cdot 13)^2 = 1002001$, an exact integer-source density $90/1002001$. In contrast, allowing *all* higher odd splitting valuations under profinite Haar gives the larger simultaneous probability $1/(16 \cdot 24 \cdot 28) = 1/10752$. The ordinary integer source $\lambda_5 = 774982$ is one explicit CRT witness. These finite certificates illustrate, but do not establish, the infinite-prime probability theorems.

19 Relation to the full CCR-k theory

A separate pre-existing part of the CCR-k theory produces an infinite unit lower-Hessenberg family $H^{[k]}_{p,\boldsymbol{\lambda}}$ and monic principal-determinant polynomials $Q_n(x) = \det(xI - H_n)$. It satisfies a universal p-blind shadow $Q_n(x) \equiv \prod_{j=1}^{n}(x-j) \pmod{p}$ and a first-visibility relation $Q_r^{\boldsymbol{\lambda}} - Q_r^{\boldsymbol{\mu}} = p(r-1)!(\lambda_r - \mu_r)$ for parameters first differing at order r; these are *prior CCR results*, not the novel theorems of this paper. General banded Hessenberg/production-matrix and d-orthogonal correspondences are well established, in particular in [10, 11]. The present $G_{p,\lambda}(m)$ is a normalized fourth-column object, not the same as the characteristic polynomial Q_4. The odd-column construction in Section 8 continues the normalized-column side, still distinct from Q_n.

20 Finite certificates and scope

The explicit congruence and resultant certificates in this work use exact integer arithmetic. For the quartic at 419, the polynomial F in (6) and its exact discriminant factor in (8) reduce the classification to the valuation and leading unit of the nonzero integer $F(\lambda)$. Irreducibility follows from the cubic reduction modulo 23; the unique exceptional 419-adic residue is identified by the linear reduction modulo 419. The finite-precision splitting classes are counted in Proposition 6.2, and the critical-depth root counts follow from Theorem 7.1.

At 73, the two derivative-critical centers are obtained by Hensel lifting in their respective residue disks. Their ordered difference obeys the unit-slope congruence in Theorem 10.1, giving an isometric parameter coordinate. The resulting simultaneous splitting and second-precision residue counts are recorded in Proposition 10.3. The source-obstruction index is the fixed divisor of the integer-valued polynomial in the proof of Proposition 12.3; its prime valuations follow from the stated polynomial congruences and give a formula for all odd orders.

The polynomial factor degrees and discriminants in the seventh- and ninth-column analysis are exact finite-field and resultant certificates. Their use in prime-density statements additionally requires the stated Galois-group arguments and the classical Chebotarev theorem. The Haar, multicolumn and adelic assertions depend instead on the explicit triangular change of p-adic source coordinates, the local quadratic valuation law, and independence across distinct primes. Finite-level examples do not substitute for these general arguments.

All results concern the stationary normalized CCR polynomial family and its formally specified integer or p-adic source spaces. They do not assert physical realizability of arbitrary source choices by chronological pushdown automata, or classify all nonstationary raw source data.

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